

Air Pollution PYQs

+Noise Pollution

Last 3 Years Detailed PYQ Solution (2022-2024)

2 SEPTEMBER 2024

Which of the following are components of NOy species in atmosphere?

A. HNO_3

B. $ClONO_2$

C. $HOONO_2$

D. NO_3

E. N_2O_5

Choose the most appropriate answer from the options given below:

1. D and E Only

2. A, C and E Only

3. A, B and C Only

4. A. B, C, D and E

ANSWER: D

NOy species in the atmosphere refer to various nitrogen oxides and related compounds that collectively represent the reactive nitrogen oxides. These include both nitrogen oxides (NO and NO_2) and more complex species. Components of NOy species typically include:

- **A. HNO_3 (Nitric acid)**
- **B. $ClONO_2$ (Chlorine nitrate)**
- **C. $HOONO_2$ (Peroxynitric acid)**
- **D. NO_3 (Nitrate radical)**
- **E. N_2O_5 (Dinitrogen pentoxide)**

All of the above listed compounds are indeed part of the NOy family.

Therefore, the correct answer is:

4. **A, B, C, D and E**

Let's dive into the different categories of nitrogen oxides: NOx, NOy, and NOz.

NOx (Nitrogen Oxides):

NOx is a general term for the primary nitrogen oxides: nitrogen monoxide (NO) and nitrogen dioxide (NO_2). These compounds are significant pollutants and play a crucial role in atmospheric chemistry.

- **Sources**: NOx emissions primarily come from combustion processes, including vehicle engines, power plants, industrial processes, and biomass burning.
- **Environmental Impact**: NOx contributes to the formation of ground-level ozone (smog), acid rain, and fine particulate matter. It also has adverse effects on human health, including respiratory problems and lung diseases.
- **Chemistry**:
    - NO: Nitrogen monoxide or nitric oxide.
    - NO_2: Nitrogen dioxide, a reactive and toxic gas.

NOy (Reactive Nitrogen Oxides):

NOy encompasses a broader group of reactive nitrogen compounds, including NO and NO_2, as well as their oxidation products and other related species. NOy is often used to represent the total reactive nitrogen in the atmosphere.

- **Components**: NOy includes NOx (NO and NO_2) plus other reactive nitrogen species such as:
    - HNO_3 (Nitric acid)
    - N_2O_5 (Dinitrogen pentoxide)
    - NO_3 (Nitrate radical)
    - $ClONO_2$ (Chlorine nitrate)
    - $HOONO_2$ (Peroxynitric acid)
- **Role in Atmosphere**: NOy species participate in various chemical reactions that influence atmospheric composition, including the formation of ozone

and secondary organic aerosols. They also play a role in nitrogen deposition to ecosystems.

NOz (Secondary Nitrogen Oxides):

NOz refers to the subset of NOy species that are produced from the oxidation of NOx, excluding NO and NO_2. It represents the secondary reactive nitrogen compounds formed in the atmosphere.

- **Components**: NOz includes species such as:
  - HNO_3 (Nitric acid)
  - N_2O_5 (Dinitrogen pentoxide)
  - NO_3 (Nitrate radical)
  - $ClONO_2$ (Chlorine nitrate)
  - $HOONO_2$ (Peroxynitric acid)
- **Formation**: NOz species are formed through photochemical reactions and oxidation processes involving NOx. These reactions typically occur in the presence of sunlight and other atmospheric constituents.
- **Significance**: NOz species contribute to the formation of secondary pollutants, such as particulate nitrates, which impact air quality and human health. They also play a role in long-range transport of reactive nitrogen and nitrogen deposition.

Summary Table

Category	Description	Components
NOx	Primary nitrogen oxides	NO, NO_2
NOy	Total reactive nitrogen oxides	NO, NO_2, HNO_3, N_2O_5, NO_3, $ClONO_2$, $HOONO_2$, etc.
NOz	Secondary nitrogen oxides (excluding NO and NO_2)	HNO_3, N_2O_5, NO_3, $ClONO_2$, $HOONO_2$, etc.

Match the List-I with List-II

LIST-I Conditions		**LIST-II** Plume Behavior	
A.	Inversion below, lapse aloft	**I.**	Fumigation
B.	Strong lapse condition	**II.**	Lofting
C.	Lapse below, Inversion aloft	**III.**	Looping
D.	Weak lapse below; Inversion aloft	**IV.**	Trapping

Choose the correct answer from the options given below:

1. A-III, B-II, C-I, D-IV
2. A-II, B-III, C-I, D-IV
3. A-II, B-I, C-III, D-IV
4. A-II, B-III, C-IV, D-I

ANSWER: B

Plume Behaviors:

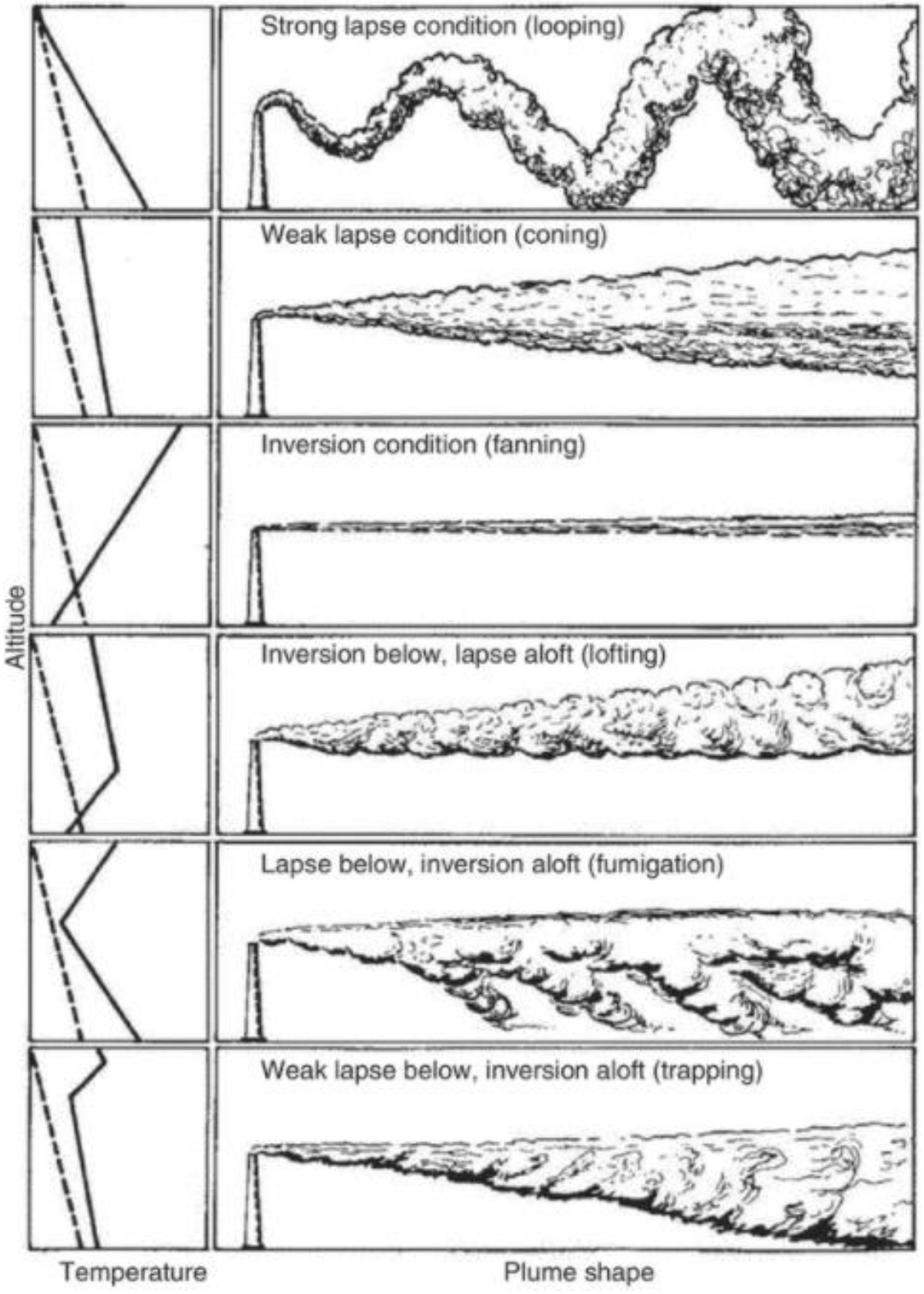
1. **Fumigation**: Occurs when an inversion layer is present below, and the air above is unstable (lapse condition aloft). This traps pollutants near the ground.
2. **Lofting**: Occurs when there is a lapse condition below (unstable air), and an inversion layer is present aloft. Pollutants are lofted upward and dispersed effectively.
3. **Looping**: Happens under strong lapse conditions, leading to vigorous vertical air mixing and causing the plume to loop.
4. **Trapping**: Occurs when there is a weak lapse condition below an inversion layer aloft, effectively trapping pollutants within a confined layer.

Matching:

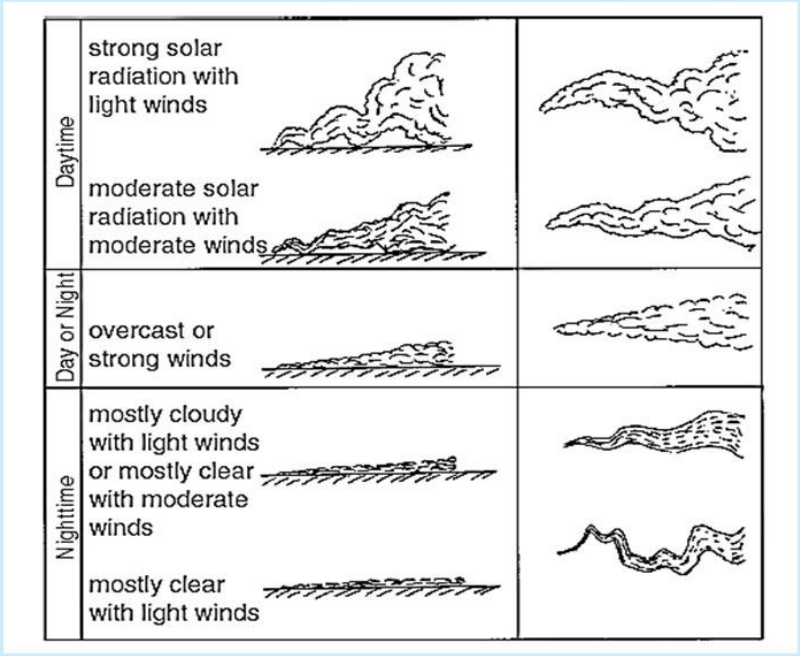
A. **Inversion below, lapse aloft**: **Fumigation** (I) B. **Strong lapse condition**: **Looping** (III) C. **Lapse below, Inversion aloft**: **Lofting** (II) D. **Weak lapse below; Inversion aloft**: **Trapping** (IV)

So, the correct answer is:

2. **A-II, B-III, C-I, D-IV**



STABLE NEUTRAL UNSTABLE



Excellent (EX)

Good (G)

Moderately Good (MG)

Moderately Poor (MP)

Poor (P)

Very Poor (VP)

Which of the following air pollution-control/particle-collection devices uses the force of gravity to aid in collection of particles by exploiting their terminal velocity?

1. Electrostatic precipitators
2. Elutriators
3. Bag houses
4. Super saturators

ANSWER: B

The air pollution-control/particle-collection device that uses the force of gravity to aid in the collection of particles by exploiting their terminal velocity is an **elutriator**.

Explanation:

- **Elutriators**: These devices separate particles based on their terminal velocities using an upward flow of air or gas. Heavier particles with higher terminal velocities settle down due to gravity, while lighter particles are carried away by the upward flow.
- **Electrostatic Precipitators**: Use electrostatic forces to collect particles from exhaust gases.
- **Bag Houses**: Use fabric filter bags to capture particles from the air.
- **Super Saturators**: Typically refer to processes that involve supersaturation conditions for condensation particle counters, not gravity-based separation.

So, the correct answer is:

2. **Elutriators**

Let's delve into the details of elutriators, a type of particle-collection device used in air pollution control and particle separation processes.

What is an Elutriator?

An elutriator is a device that separates particles based on their size, shape, and density using the principle of terminal velocity and the force of gravity. It leverages an upward flow of air or gas to classify particles, causing heavier particles to settle and lighter particles to be carried away.

How Does an Elutriator Work?

1. **Airflow Introduction**:
    - An upward flow of air or gas is introduced at the bottom of the elutriator chamber. This flow must be precisely controlled to create a consistent velocity gradient.
2. **Particle Inlet**:
    - Particles are introduced into the chamber where they encounter the upward airflow.
3. **Terminal Velocity**:
    - Each particle has a unique terminal velocity, which is the constant speed a particle reaches when the force of gravity is balanced by the drag force of the air or gas.
4. **Separation**:
    - Particles with terminal velocities greater than the upward airflow velocity will settle down and be collected at the bottom.
    - Particles with terminal velocities lower than the upward airflow velocity will be carried upward with the airflow and collected at the top or in subsequent chambers.
5. **Collection Zones**:
    - The chamber may be divided into different zones or stages to capture particles of various sizes. Each zone has a different airflow rate, allowing finer separation.

Applications of Elutriators

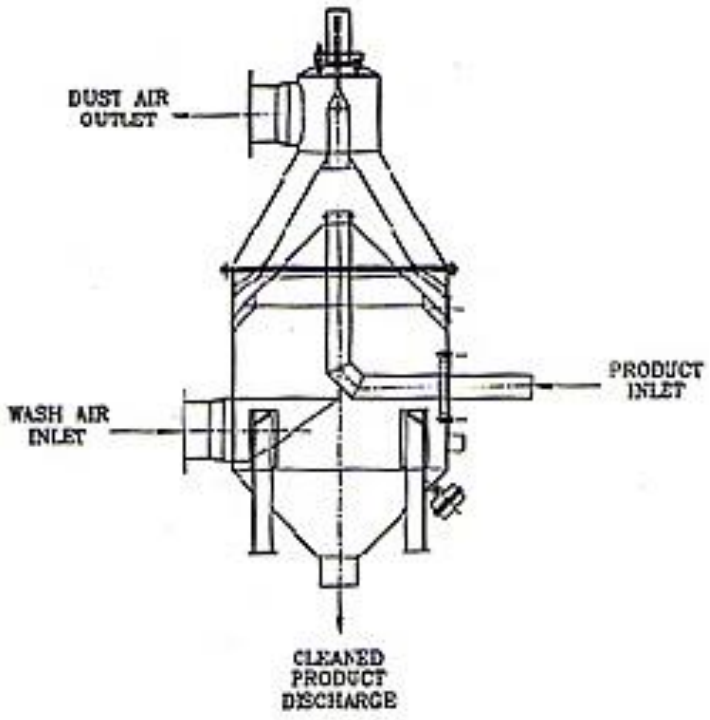
- **Air Pollution Control**: Used to remove particulate matter from industrial emissions.
- **Particle Size Classification**: Employed in industries such as pharmaceuticals, chemicals, and mining to classify particles by size.
- **Research and Laboratory Analysis**: Utilized in scientific studies to analyze particle size distributions and aerodynamic properties.

Advantages of Elutriators

- **Simple Design**: Elutriators have a straightforward design with no moving parts, making them easy to operate and maintain.
- **Effective Separation**: Capable of separating particles with high precision based on their terminal velocity.
- **Versatility**: Can be used for a wide range of particle sizes and densities.

Disadvantages of Elutriators

- **Limited Capacity**: Not suitable for handling large volumes of particles.
- **Airflow Control**: Requires precise control of airflow to ensure accurate separation.



Choose the correct statements about ozone in stratosphere.

A. Rate of O_3 production is highest at equator and at about 40 km height.

B. Maximum O_3 concentration at equator occur at about 25 km as opposed to 40 km height.

C. At poles, maximum O_3 production occur at altitude even higher than those over equator.

D. Regions of highest O_3 concentration do not coincide with location of highest rate of O_3 formation

E. Catalytic destruction of O_3 is more at Artic compared to Antarctica due to more CFCs in northern hemisphere.

Choose the most appropriate answer from the options given below:

1. A, C and D Only

2. B, D and E Only

3. A, B, C and D Only

4. A, D and E Only

ANSWER: C

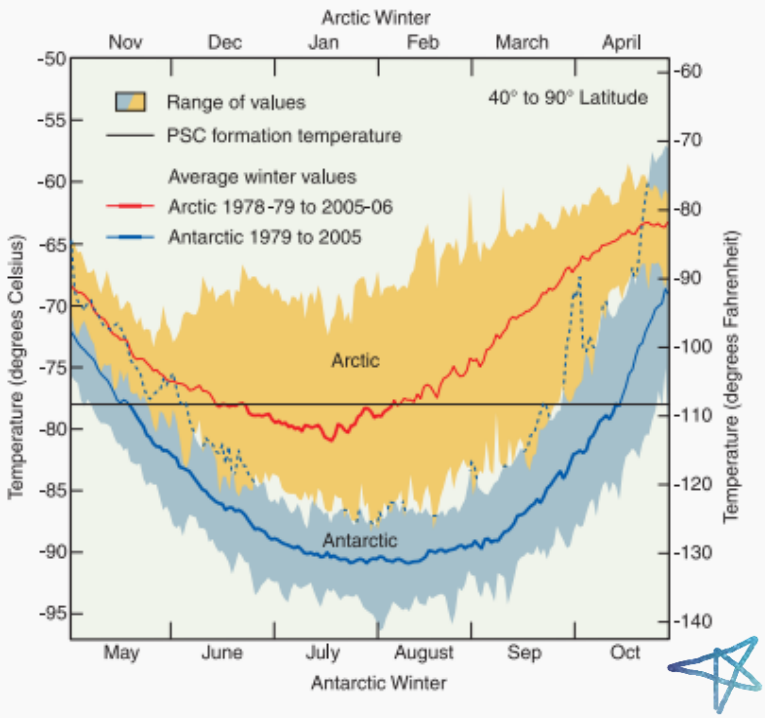


Figure Q10-1. Arctic and Antarctic temperatures. Stratospheric air temperatures in both polar regions reach minimum values in the lower stratosphere in the winter season. Average minimum values over Antarctica are as low as –90°C in July and August in a typical year. Over the Arctic, average minimum values are near –80°C in January and February. Polar stratospheric clouds (PSCs) are formed when winter minimum temperatures fall below the formation temperature (about –78°C). This occurs on average for 1 to 2 months over the Arctic and 5 to 6 months over Antarctica (see heavy red and blue lines). Reactions on PSCs cause the highly reactive chlorine gas ClO to be formed, which increases the destruction of ozone (see Q9). The range of winter minimum temperatures found in the Arctic is much greater than in the Antarctic. In some years, PSC formation temperatures are not reached in the Arctic, and significant ozone depletion does not occur. In the Antarctic, PSCs are present for many months, and severe ozone depletion now occurs in each winter season.

Noise coming out during dentists drilling would be an appropriate example of

1. Steady state noise
2. Intermittent noise
3. Impulse noise
4. Impact noise

ANSWER: B

The noise produced during a dentist's drilling is best categorized as **intermittent noise**.

Explanation:

- **Steady State Noise**: Continuous noise that remains relatively constant over time. Examples include the hum of an air conditioner or a running fan.
- **Intermittent Noise**: Noise that starts and stops at intervals. The sound from a dentist's drill fits this category as it is not constant but occurs in bursts, starting and stopping as the dentist operates the drill.
- **Impulse Noise**: Noise that occurs as a sudden burst or spike, such as a gunshot or a balloon popping. This type of noise is short and sharp.
- **Impact Noise**: Similar to impulse noise, it is caused by collision events, like hammering or banging.

So, the correct answer is:

2. **Intermittent noise**

Summary Table

Type of Noise	Definition	Examples	Characteristics
Steady State Noise	Constant over time	Air conditioner hum, office noise	Blends into background, less noticeable
Intermittent Noise	Occurs at intervals, sporadic	Dentist's drill, traffic noise	Noticeable, associated with specific events
Impulse Noise	Short bursts, sudden onset	Gunshot, balloon popping	Startling, can cause discomfort
Impact Noise	Results from collision/impact	Hammering, dropping objects	Varies in intensity, physical impacts
Broadband Noise	Contains a wide range of frequencies	White noise, waterfall sound	Soothing, used in sound masking
Narrowband Noise	Confined to a narrow range of frequencies	Electrical hum, tuning fork	Intrusive, specific frequency
Environmental Noise	Originates from environment, natural and human-made sources	Birds chirping, traffic, construction	Varies widely, impacts quality of life

Which one of the following is a greenhouse gas and a criteria air pollutant under national ambient air quality standards but is not directly emitted in troposphere?

1. N_2O
2. CO_2
3. SF_6
4. O_3

ANSWER: D

Explanation:

- **Ozone (O_3)**:
  - **Greenhouse Gas**: Ozone is a greenhouse gas, contributing to the greenhouse effect and global warming.
  - **Criteria Air Pollutant**: It is also one of the six criteria air pollutants regulated under the National Ambient Air Quality Standards (NAAQS) due to its harmful effects on human health and the environment.
  - **Not Directly Emitted**: Ozone in the troposphere is not directly emitted. Instead, it is formed through photochemical reactions involving precursor pollutants like nitrogen oxides (NOx) and volatile organic compounds (VOCs) in the presence of sunlight.

Other Options:

- **N_2O (Nitrous Oxide)**: A greenhouse gas but not a criteria air pollutant.
- **CO_2 (Carbon Dioxide)**: A greenhouse gas but not a criteria air pollutant under NAAQS.
- **SF_6 (Sulfur Hexafluoride)**: A potent greenhouse gas but not a criteria air pollutant.

So, the correct answer is:

4. **O_3 (Ozone)**

The essential conditions for photochemical smog formation in ambient atmosphere are

1. SO_2, water vapor, T>25° C
2. Smoke, water vapor, T>25° C
3. VOC, SO_2, T>25° C
4. NOx, VOC,T>25° C

ANSWER: D

The essential conditions for photochemical smog formation in the ambient atmosphere are:

4. **NO, VOC, T > 25°C**

Explanation:

- **NO (Nitric Oxide)**: Nitrogen oxides (NOx), primarily NO and NO_2, are key precursors in the formation of photochemical smog. They are emitted from vehicle exhaust, industrial processes, and other combustion sources.
- **VOC (Volatile Organic Compounds)**: VOCs are organic chemicals that easily evaporate and enter the atmosphere. They come from sources like vehicle exhaust, industrial emissions, and the use of solvents and paints.
- **T > 25°C**: Higher temperatures, usually above 25°C, provide the necessary energy for the photochemical reactions to occur. Sunlight is a crucial factor, as it drives the chemical reactions that lead to the formation of smog.

Photochemical smog, also known as "ozone smog," is formed through a complex series of photochemical reactions involving NOx and VOCs in the presence of sunlight, leading to the production of ground-level ozone (O_3) and other harmful pollutants.

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Arrange the following atmospheric species in increasing order of their residence times :

(A) Formaldehyde

(B) Hydroxyl radical

(C) Carbon monoxide

(D) Nitrous oxide

Choose the correct answer from the options given below :

(1) (B), (A), (C), (D)

(2) (A), (B), (C), (D)

(3) (B), (D), (A), (C)

(4) (D), (B), (A), (C)

ANSWER: A

To arrange the atmospheric species in increasing order of their residence times, we need to understand their typical lifetimes in the atmosphere:

- **Hydroxyl radical (OH)**: Extremely reactive with a very short lifetime, typically on the order of seconds to minutes.
- **Formaldehyde (HCHO)**: Short-lived, with a lifetime ranging from hours to days.
- **Carbon monoxide (CO)**: Moderately reactive, with a lifetime of weeks to months.
- **Nitrous oxide (N_2O)**: Long-lived, with a lifetime of over 100 years.

Arranging in Increasing Order of Residence Times:

1. **Hydroxyl radical (B)**: Seconds to minutes.
2. **Formaldehyde (A)**: Hours to days.

3. **Carbon monoxide (C)**: Weeks to months.
4. **Nitrous oxide (D)**: Over 100 years.

Therefore, the correct order is:

1. **(B), (A), (C), (D)**

Species	Type	Approx. Atmospheric Lifetime	Global Warming Potential (GWP)
Carbon Dioxide (CO_2)	Greenhouse Gas	100 - 300 years	1 (Baseline GWP)
Methane (CH_4)	Greenhouse Gas	~12 years	86 (20 years)
Nitrous Oxide (N_2O)	Greenhouse Gas	~114 years	268 (20 years)
CFC-12	Greenhouse Gas	~100 years	10,200 (20 years)
HFC-134a	Greenhouse Gas	~14 years	3,710 (20 years)
Sulfur Hexafluoride (SF_6)	Greenhouse Gas	~3,200 years	17,500 (20 years)
Hydroxyl Radical (OH)	Pollutant	Seconds to minutes	N/A
Formaldehyde (HCHO)	Pollutant	Hours to days	N/A
Carbon Monoxide (CO)	Pollutant	Weeks to months	N/A
Ozone (O_3) in Troposphere	Pollutant	Hours to weeks	N/A

Which of the following indices of noise pollution would be most appropriate for assessing the background noise level at a place?

(1) L_{90}

(2) L_{50}

(3) L_{eq}

(4) L_{10}

ANSWER: A

Let's delve into the various indices of noise pollution and understand each one in detail:

Noise Pollution Indices

1. **L10**
2. **L50**
3. **L90**
4. **Leq**

Detailed Explanation:

1. L10

- **Definition**: L10 is the noise level that is exceeded 10% of the time during a measurement period.
- **Usage**: It is often used to assess higher noise levels, such as those from traffic or other intermittent noise sources.
- **Significance**: It indicates the peak noise level that is commonly experienced and is useful for understanding the impact of high noise events on the environment and human health.

2. L50

- **Definition**: L50 represents the noise level that is exceeded 50% of the time during a measurement period.

- **Usage**: It provides an indication of the median or typical noise level in an area.
- **Significance**: L50 is useful for giving a general sense of the noise environment and is less affected by extreme noise events compared to L10.

3. L90

- **Definition**: L90 is the noise level that is exceeded 90% of the time during a measurement period.
- **Usage**: It is commonly used to assess background noise levels.
- **Significance**: L90 effectively filters out short-term noise peaks and focuses on the more consistent, underlying noise level, making it ideal for assessing background noise.

4. Leq (Equivalent Continuous Sound Level)

- **Definition**: Leq is the equivalent continuous sound level over a specified period. It represents the constant sound level that contains the same acoustic energy as the actual varying sound levels over the measurement period.
- **Usage**: Leq is widely used to assess overall noise exposure in various environments, including urban areas, workplaces, and industrial settings.
- **Significance**: Leq takes into account all fluctuations in noise levels, providing a comprehensive measure of noise exposure over time.

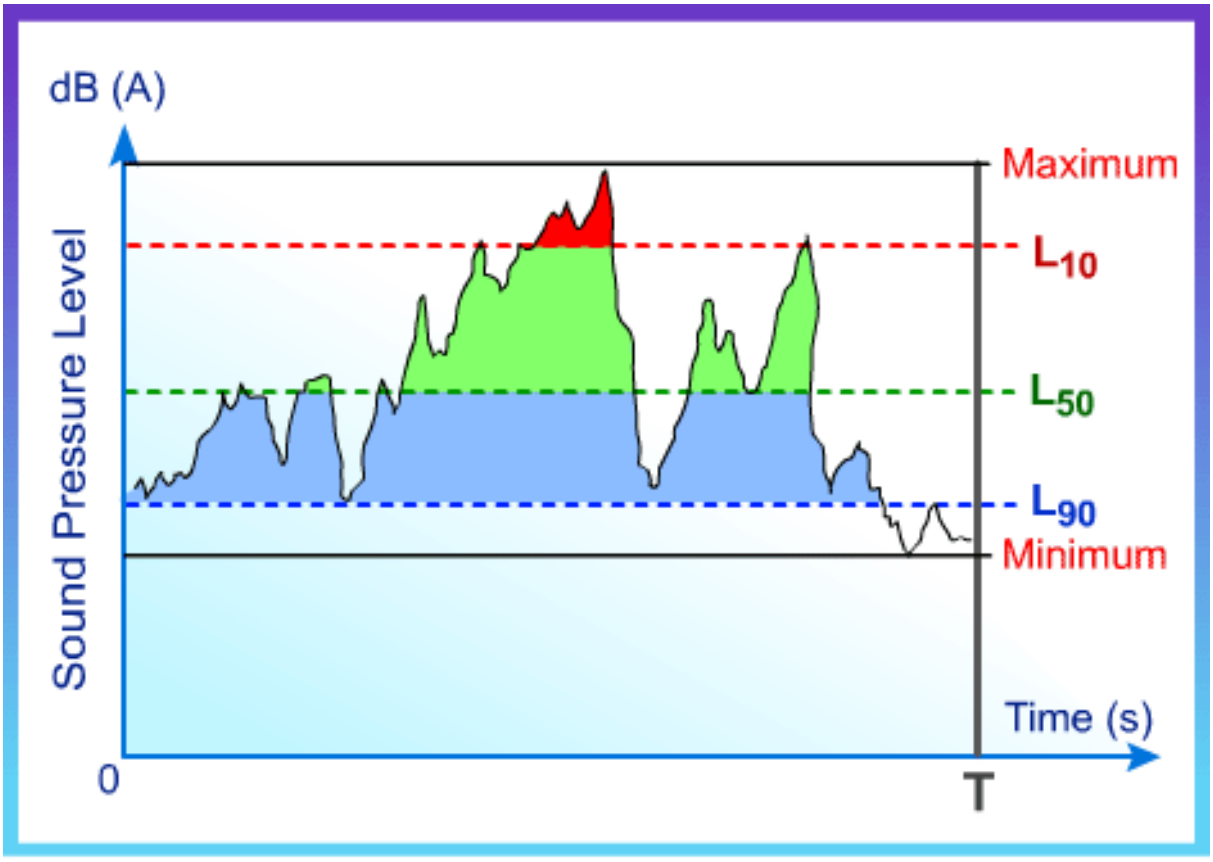
Example Scenario

Imagine you are assessing noise pollution in a residential area near a busy road. Here's how you might use these indices:

- **L10**: Measures the noise level exceeded during the loudest moments (e.g., when heavy traffic passes).
- **L50**: Provides a median noise level, giving an idea of the typical noise environment.
- **L90**: Focuses on background noise, useful for understanding the baseline noise level when traffic is low or absent.
- **Leq**: Combines all noise levels over a period to provide an overall assessment of noise exposure.

Summary Table

Index	Definition	Usage	Significance
L10	Noise level exceeded 10% of the time	Assessing peak noise levels	Indicates the highest noise events
L50	Noise level exceeded 50% of the time	General noise environment assessment	Provides median noise level
L90	Noise level exceeded 90% of the time	Assessing background noise levels	Focuses on consistent, underlying noise
Leq	Equivalent continuous sound level	Overall noise exposure assessment	Comprehensive measure of noise exposure



Please note that $L_{10} > L_{50} > L_{90}$ for the same sound or noise.

Dobson unit is the thickness of Ozone which is expressed in :

(1) Tenths of centimetre

(2) Tenths of millimetre

(3) Hundredths of millimetre

(4) Hundredths of centimetre

ANSWER: A

A Dobson Unit (DU) is a measure of the total amount of ozone in a column of the atmosphere, and it is defined as the thickness of the ozone layer that would be formed if it were compressed to standard temperature and pressure (STP). One Dobson Unit corresponds to a thickness of:

3. **Hundredths of millimetre**

Average Dobson Units:

1. **Normal Conditions**:
   - **Average Range**: Typically, the total column ozone amount varies between 250 to 500 Dobson Units (DU).
   - **Example**: In mid-latitude regions, an average ozone column might be around 300 to 350 DU.
2. **Ozone Hole Conditions**:
   - **Average Range**: During the annual occurrence of the ozone hole over Antarctica, the ozone concentration can drop significantly.
   - **Example**: In severe cases, ozone levels can fall below 220 DU, which is used as a benchmark for defining the ozone hole. In the worst instances, it can drop as low as 100 DU or even less.

Condition	Typical Dobson Units (DU)
Normal Conditions	250 - 500 DU
Ozone Hole Conditions	< 220 DU
Severe Ozone Hole	As low as 100 DU or less

Of the following species, the most likely to be found in reducing smog is :

(1) Ozone

(2) SO_2

(3) Peroxyacetyl nitrate

(4) Peroxybenzoyl nitrate

ANSWER: B

Reducing smog, also known as **London-type smog** or **industrial smog**, is primarily associated with sulfur dioxide (SO_2) and particulate matter. It typically forms in cool, humid conditions and is linked to the burning of coal and other fossil fuels.

The species most likely to be found in reducing smog is:

2. **SO_2 (Sulfur Dioxide)**

In a region where radiation inversions extend beyond the stack height, the emitted plume will be best characterized as:

(1) Coning

(2) Fanning

(3) Trapping

(4) Looping

ANSWER: B

In a region where radiation inversions extend beyond the stack height, the emitted plume will be best characterized as:

2. **Fanning**

Explanation:

- **Fanning**: This plume behavior occurs when there is a temperature inversion in the atmosphere, causing a stable layer that prevents vertical mixing. The emitted pollutants spread horizontally, creating a fanning pattern. This typically happens during radiation inversions, where cool air is trapped near the ground under a layer of warmer air, extending above the stack height.

Here's a quick overview of the other plume behaviors for comparison:

- **Coning**: Occurs in neutral stability conditions, where pollutants disperse evenly in a cone-shaped pattern.
- **Trapping**: Happens when there are inversions both above and below the emission source, trapping pollutants in a layer.
- **Looping**: Occurs in highly unstable conditions with strong vertical mixing, causing the plume to loop.

For each 10°C rise in temperature, the rate of biological processes roughly increases by:

(1) 2 times

(2) 3 times

(3) 4 times

(4) 5 times

ANSWER: A

For each 10°C rise in temperature, the rate of biological processes roughly increases by:

1. **2 times**

Explanation:

This concept is commonly referred to as the **Q_{10} temperature coefficient**, which describes the rate of increase in biological or chemical processes with a 10°C rise in temperature. For most biological processes, the Q_{10} value is typically around 2, meaning that the rate of these processes doubles with every 10°C increase in temperature.

Which of the following are correct statements ?

(A) Potential temperature is a conserved quantity for an air parcel under adiabatic condition.

(B) The air parcel temperature becomes equal to dew point temperature at lifting condensation level.

(C)The saturation vapour pressure over pure water increases linearly with temperature of water.

(D) The magnitude of saturated adiabatic lapse rate is always greater than that of dry adiabatic lapse rate.

Choose the correct answer from the options given below:

(1) (A), (B) and (C) Only

(2) (A) and (B) Only

(3) (B), (C) and (D) Only

(4) (A), (B) and (D) Only

ANSWER: B

Let's evaluate each statement:

(A) **Potential temperature is a conserved quantity for an air parcel under adiabatic condition.**

- This statement is **true**. Potential temperature remains constant for an air parcel undergoing adiabatic processes (no heat exchange with the environment).

(B) **The air parcel temperature becomes equal to dew point temperature at lifting condensation level.**

- This statement is **true**. At the lifting condensation level (LCL), the temperature of the air parcel cools to its dew point temperature, leading to condensation.

(C) **The saturation vapour pressure over pure water increases linearly with temperature of water.**

- This statement is **false**. The saturation vapour pressure over water increases exponentially with temperature, not linearly.

(D) **The magnitude of saturated adiabatic lapse rate is always greater than that of dry adiabatic lapse rate.**

- This statement is **false**. The saturated adiabatic lapse rate is typically less than the dry adiabatic lapse rate because the release of latent heat during condensation reduces the rate of cooling.

Given this analysis, the correct answer is:

2. **(A) and (B) Only**

Which of the following devices are used for controlling the particulate contaminants present in stack emissions?

(A) Wet Collectors

(B) Centrifugal Collectors

(C) Electrostatic Precipitators

(D) Baghouse filters

(E) Fixed bed adsorbers

Choose the correct answer from the options given below:

(1) (A), (B) and (C) Only

(2) (B), (C) and (D) Only

(3) (A), (B), (C) and (D) Only

(4) (A), (B), (C), (D) and (E)

ANSWER: C

The devices used for controlling particulate contaminants in stack emissions are:

(A) Wet Collectors (B) Centrifugal Collectors (C) Electrostatic Precipitators (D) Baghouse filters

Fixed bed adsorbers (E) are typically used for controlling gaseous contaminants, not particulate matter.

Therefore, the correct answer is:

3. **(A), (B), (C) and (D) Only**

Types of Pollution Control Devices

Type	Technology/Device	Description
Particulate Contaminant Control		
Wet Collectors	Wet Scrubbers	Use liquid to remove particles from exhaust gas.
Centrifugal Collectors	Cyclone Separators	Use centrifugal force to separate particles from gas stream.
Electrostatic Precipitators	Electrostatic Precipitators (ESPs)	Use electrical charges to attract and remove particles from gas stream.
Baghouse Filters	Fabric Filters	Use fabric bags to capture particles from exhaust gas.
Mechanical Collectors	Settling Chambers	Use gravity to settle particles from gas stream.
High-Efficiency Particulate Air (HEPA) Filters	HEPA Filters	Use fine mesh to capture very small particles from air.
Gaseous Contaminant Control		
Absorption	Packed Bed Absorbers	Use a liquid solvent to absorb gaseous pollutants from exhaust gas.
Adsorption	Activated Carbon Adsorbers	Use porous materials to adsorb gaseous pollutants from gas stream.
Combustion	Thermal Oxidizers	Use high temperatures to combust and remove gaseous pollutants.
Catalytic Combustion	Catalytic Converters	Use catalysts to promote the combustion of gaseous pollutants at lower temperatures.
Biofiltration	Biofilters	Use microorganisms to degrade gaseous pollutants as the gas passes through a biologically active filter bed.

Vehicular Pollution Control		
Catalytic Converters	Three-Way Catalytic Converters	Reduce NOx, CO, and unburned hydrocarbons in vehicle exhaust.
Diesel Particulate Filters (DPFs)	Diesel Particulate Filters	Capture soot and particulate matter from diesel engine exhaust.
Exhaust Gas Recirculation (EGR)	EGR Systems	Reduce NOx emissions by recirculating a portion of the exhaust gas back into the engine cylinders.
Selective Catalytic Reduction (SCR)	SCR Systems	Use a reductant (like urea) and a catalyst to reduce NOx emissions in diesel engines.
Evaporative Emission Control	Evaporative Emission Control Systems	Prevent fuel vapors from escaping into the atmosphere from the fuel system.
On-Board Diagnostics (OBD)	OBD Systems	Monitor and report vehicle emission control system performance and malfunctions.
Advanced Engine Technologies	Hybrid and Electric Engines	Reduce emissions by using alternative power sources and improving fuel efficiency.

Identify the conditions which are favourable for the formation of urban heat islands:

(A) Clear sky

(B) High albedo over urban areas

(C) Low evapo-transpiration over urban areas

(D) High wind speed

Choose the correct answer from the options given below:

(1) (A), (B) and (D) Only

(2) (A) and (C) Only

(3) (B) and (C) Only

(4) (B) and (D) Only

ANSWER: B

Let's evaluate each condition for the formation of urban heat islands:

(A) **Clear sky**: True. Clear skies allow for maximum solar radiation to reach the urban surfaces, increasing temperatures.

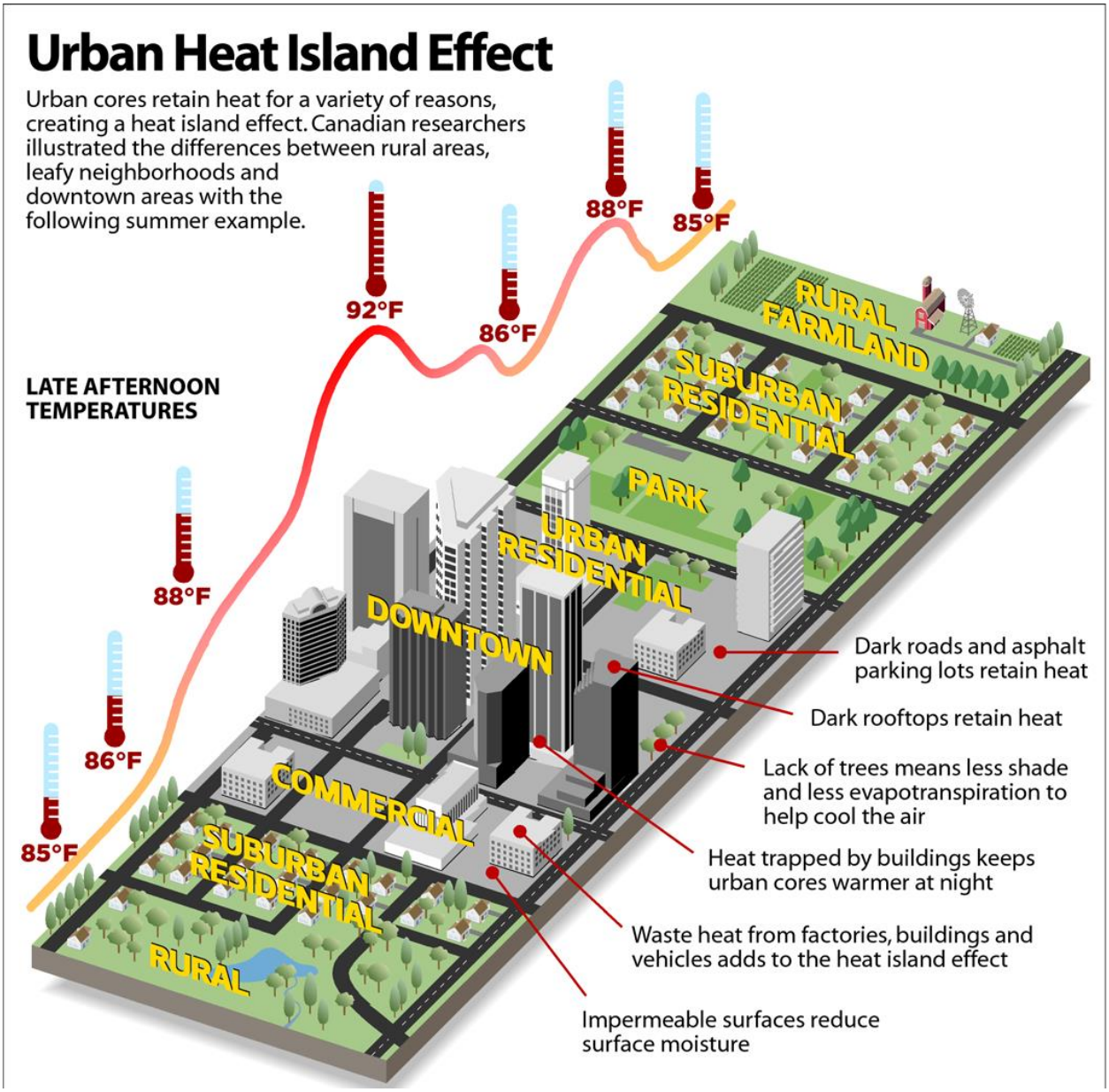
(B) **High albedo over urban areas**: False. High albedo (reflectivity) means surfaces reflect more solar radiation, which would reduce heat absorption and thus not favor urban heat island formation. Typically, urban areas have low albedo.

(C) **Low evapo-transpiration over urban areas**: True. Urban areas usually have less vegetation and water bodies, leading to lower evapotranspiration, which means less cooling.

(D) **High wind speed**: False. High wind speeds enhance mixing of air and dissipate heat, which is not favorable for urban heat island formation. Low wind speeds are more conducive to heat build-up.

Given this analysis, the correct answer is:

2. **(A) and (C) Only**



Identify Trihalomethanes (THMs) among the following:

(A) chloroform

(B) bromodichloromethane

(C) dibromochloromethane

(D) bromoform

Choose the correct answer from the options given below :

(1) (A) Only

(2) (D) Only

(3) (B) and (C) Only

(4) (A), (B), (C) and (D)

ANSWER: D

Trihalomethanes (THMs) are chemical compounds that consist of three halogen atoms (chlorine, bromine, or iodine) substituted for three of the hydrogen atoms in methane (CH_4). The following compounds are identified as THMs:

- (A) Chloroform ($CHCl_3$)
- (B) Bromodichloromethane ($CHBrCl_2$)
- (C) Dibromochloromethane ($CHBr_2Cl$)
- (D) Bromoform ($CHBr_3$)

Therefore, the correct answer is:

4. **(A), (B), (C) and (D)**

Which of the following are used to control SO_2 emissions from a thermal power plant ?

(A) Aqueous solutions of sodium and ammonia

(B) Liquid nitrogen

(C) Aqueous solutions of calcium and magnesium

(D) Chlorine

Choose the correct answer from the options given below :

(1) (A) and (B) Only

(2) (A) Only

(3) (C) and (D) Only

(4) (A) and (C) Only

ANSWER: D

The substances used to control SO_2 emissions from a thermal power plant typically involve chemical reactions that neutralize the sulfur dioxide. The correct choices are:

- **Aqueous solutions of sodium and ammonia** (A)
- **Aqueous solutions of calcium and magnesium** (C)

Both options (A) and (C) are commonly used in wet scrubbers and flue gas desulfurization (FGD) systems to remove SO_2 from the emissions by forming compounds like sodium sulfite, sodium bisulfite, calcium sulfite, and calcium sulfate.

Therefore, the correct answer is:

4. **(A) and (C) Only**

JUNE 2024 Cancelled

Noise level of 90 dBA exists for 10 minutes and is followed by reduced noise level of 70 dBA for 40 minutes. Calculate the equivalent continuous equal energy level for 50-minute period. Assume 5-minute sampling interval:

(1) ~32 dBA

(2) ~64 dBA

(3) ~83 dBA

(4) -160 dBA

ANSWER: C

The right answer will be 86.16 DbA

Formula:

$$Leq = 10 \cdot \log_{10}\left(\frac{1}{T}\sum_{i=1}^{n} t_i \cdot 10^{\frac{L_i}{10}}\right)$$

Steps

1. Convert each noise level to its corresponding power ratio

2. Multiply each power ratio by its respective duration

3. Sum these values and divide by total time period

4. Take the logarithm and multiply by 10 to find Leq

Answer: 86.16 Dba

What would be the behaviour of smokestack plume under the strong lapse rate condition in which the decrease in temperature vs. altitude is much greater than normal?

(1) Trapping

(2) Fumigation

(3) Looping

(4) Conning

ANSWER: C

Under a strong lapse rate condition where the decrease in temperature with altitude is much greater than normal, the behavior of a smokestack plume would be **(3) Looping**. This occurs because the environmental lapse rate exceeds the dry adiabatic lapse rate, creating instability and causing the plume to display a looping tendency—rising, falling, and then rising again

The normal lapse rate of temperature per kilometer is

(1) 9.6°

(2) 10.6° C

(3) 6.4° C

(4) 4.6° C

ANSWER: C

The normal lapse rate of temperature per kilometer is **6.4° C** (Option 3). This rate is defined by the International Civil Aviation Organization (ICAO) as an average temperature decrease of 6.50 °C/km from sea level to 11 km in altitude

The collection mechanism of a bag filter includes

A. Sedimentation

B. Diffusion

C. Electrostatic Interaction

D. Interception

E. Impaction

Choose the correct answer from the options given below:

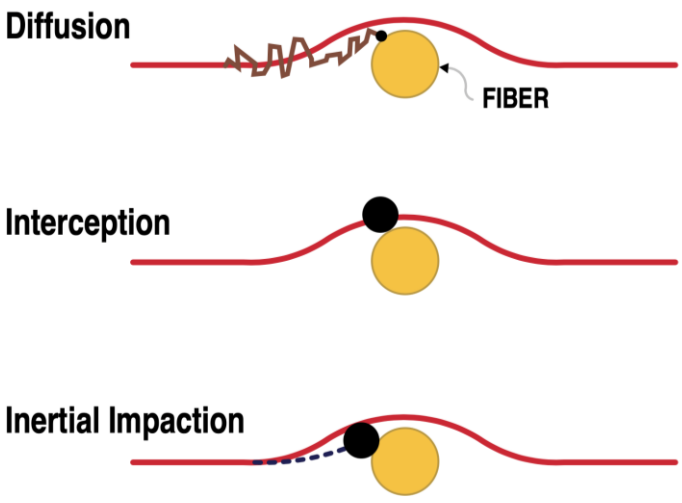
(1) A, B and D only

(2) A, B, D and E only

(3) B, D and E only

(4) A, B, C, D and E

ANSWER: C



Choose the incorrect statement:

(1) O_2 undergo photo-dissociation by UV-C radiation in stratosphere

(2) O_3 undergo photo-dissociation by UV-B radiation in stratosphere

(3) NO/NO_2 catalytic destruction cycle of ozone in stratosphere is initiated by photochemical of N_2O

(4) Catalytically active reservoir of Cl in stratosphere are HCI gas and $CIONO_2$

ANSWER: C and D

Let's analyze each statement to identify the incorrect one:

1. **O_2 undergo photo-dissociation by UV-C radiation in the stratosphere**:
    - **Correct**. Oxygen molecules (O_2) are photodissociated by UV-C radiation (wavelengths less than 240 nm) in the stratosphere, leading to the formation of atomic oxygen.
2. **O_3 undergo photo-dissociation by UV-B radiation in the stratosphere**:
    - **Correct**. Ozone (O_3) is photodissociated by UV-B radiation (wavelengths between 280 and 320 nm) in the stratosphere, resulting in the formation of molecular oxygen and atomic oxygen.
3. **NO/NO_2 catalytic destruction cycle of ozone in the stratosphere is initiated by photochemical of N_2O**:
    - **Incorrect**. The NO/NO_2 catalytic destruction cycle of ozone is initiated by the photochemical reaction of nitrogen dioxide (NO_2), not nitrous oxide (N_2O). N_2O is relatively stable and is transported to the stratosphere where it can be photolyzed to form NO, but it does not directly initiate the catalytic cycle.
4. **Catalytically active reservoirs of Cl in the stratosphere are HCl gas and $ClONO_2$**:

- **Partially Correct**. Hydrogen chloride (HCl) and chlorine nitrate ($ClONO_2$) are reservoirs for chlorine in the stratosphere. While they are reservoirs, they are generally considered inactive forms that can release active chlorine under certain conditions.

Given this analysis, the incorrect statement is:

3. **NO/NO_2 catalytic destruction cycle of ozone in the stratosphere is initiated by photochemical of N_2O**

Which one of the following pollutant is produced while burning hydrogen as fuel?

(1) CO_2

(2) NO_2

(3) H_2O

(4) SO_X

ANSWER: B

When hydrogen is burned as fuel, it can produce nitrogen oxides (NOx), including nitric oxide (NO) and nitrogen dioxide (NO_2) . These NOx emissions are harmful air pollutants. So, the correct answer is **(2) NO_2**. Unlike the common misconception that hydrogen only produces water, its combustion can indeed result in NOx emissions.

Arrange the following gases in increasing rate of diffusion during aeration process:

A. H_2

B. O_2

C. Cl_2

D. O_3

E. Bi

Choose the correct answer from the options given below:

(1) A, B, C, D, E

(2) A, B, D, C, E

(3) E, C, D, B, A

(4) E, D, C, B, A

ANSWER: D

1. **H_2 (hydrogen)**: Hydrogen gas is relatively light and has a high diffusion rate due to its small molecular size.
2. **O_2 (oxygen)**: Oxygen gas also diffuses quickly, especially since it is essential for respiration.
3. **Cl_2 (chlorine)**: Chlorine gas has a moderate diffusion rate.
4. **O_3 (ozone)**: Ozone gas has a slower diffusion rate compared to hydrogen and oxygen.
5. **Bi (bismuth)**: Bismuth gas has the slowest diffusion rate among the listed gases.

Arranging them in increasing order of diffusion rate during aeration:

1. **Bi** (slowest)
2. **O_3**
3. **Cl_2**
4. **O_2**
5. **H_2** (fastest)

Therefore, the correct order is **(4) E, D, C, B, A**.

According to the World Air Quality Report 2023 relied by the Swiss Organization IQ Air, which of the following metropolitan area is listed as the most polluted area?

(1) Begusarai

(2) Bengaluru

(3) Surat

(4) Varanasi

ANSWER: A

The **most polluted metropolitan area** in 2023, according to the **World Air Quality Report**, was **Begusarai** in Bihar

State of Global Air (SoGA) Report, 2024

- It is released by the **Health Effects Institute (HEI),** an independent **U.S.-based nonprofit** research organization, **in partnership with UNICEF.**
- The report provides an analysis of data for air quality and health impacts for countries around the world.
- It defined air pollution as a complex mixture including particles and different gases, with sources and compositions varying over space and time.
- **Highlights of SoGA Report 2024:**
    - Air pollution has become the **second-leading** global risk **factor for death,** surpassing tobacco and diabetes and **trailing only hypertension**.
    - **Among these, non communicable diseases,** including heart disease, stroke, diabetes, lung cancer, and chronic obstructive pulmonary disease, accounted for nearly **90% of the disease burden** from air pollution.

- **Six out of ten deaths** worldwide **attributed to air pollution** are **caused by the tiny PM2.5 particle**; the remaining two pollutants, ozone and household air pollution, account for 38% and 6% of deaths, respectively.
- Diseases and ailments linked to **air pollution claimed 8.1 million lives** worldwide **in 2021,** with **India accounting for one in four of these** deaths.
- **India and China together account for 54%** of the total global disease burden due to air pollution in 2021. **In India, air pollution accounted for 21 lakh deaths,** and in **China, it caused 23 lakh deaths.**
- More than 7**,00,000 deaths in children under five years** were **linked to air pollution**. This represents **15% of all global deaths among children under five.**
- Lower respiratory infections (LRIs) are the leading cause of death for children under five
- **India** recorded the **highest number of deaths among children under five due to air pollution** worldwide **in 2021.**
- Air pollution in India was responsible for the deaths of 1,69,400 children under the age of five in 2021. This means that around 464 children died every day in India that year due to diseases caused by air pollution.
- **In 2021,nearly 50%** (2,37,000 deaths) **of all ozone-related COPD** (chronic obstructive pulmonary disease) **deaths were in India**, followed by China with 1,25,600 deaths and Bangladesh with 15,000 deaths.

Which of the following is a criteria pollutant and a greenhouse gas but is not directly emitted in the ambient atmosphere ?

(1) Carbon dioxide

(2) Nitrous oxide

(3) Ozone

(4) Nitrogen dioxide

ANSWER: C

The criteria pollutant that is also a greenhouse gas but is not directly emitted into the ambient atmosphere is **ozone (O_3)** Ozone is a secondary pollutant formed through chemical reactions involving precursor pollutants like nitrogen oxides (NO_x) and volatile organic compounds (VOCs). It contributes to both air quality degradation and climate change. Nitrous oxide (N_2O) is also a greenhouse gas, but it is not considered a criteria pollutant . Carbon dioxide (CO_2) and nitrogen dioxide (NO_2) are directly emitted pollutants but are not considered greenhouse gases in the same way as ozone or nitrous oxide

December 2023

Ozone Depleting Potential (ODP) of any substance/chemical species is defined as the ratio of long term impact on ozone from substance/chemical species to the impact from an equivalent amount of

(1) CCI_4

(2) CFC-11

(3) CFC-113

(4) CFC-115

ANSWER: B

Ozone Depleting Potential (ODP) is a measure of the relative amount of ozone depletion caused by a substance compared to the impact of a reference substance. The reference substance commonly used for calculating ODP is:

2. **CFC-11 (Trichlorofluoromethane)**

Explanation:

ODP is defined as the ratio of the long-term impact on ozone from a substance/chemical species to the impact from an equivalent amount of CFC-11. CFC-11 has an ODP of 1, and other substances are measured relative to this value.

Therefore, the correct answer is:

2. **CFC-11**

Gaussian plume model for estimating pollutant concentration from point source considers the following assumptions:

(A) Concentrations are proportional to emission rate

(B) Emissions takes place from virtual point

(C) Pollutant does not undergo chemical transformations

(D) Three-dimensional axis system of downwind, crosswind and vertical with origin at the ground

(E) Considers variable wind speed and direction

Choose the correct answer from the options given below:

(1) (A), (D) and (E) Only

(2) (A), (C) and (D) Only

(3) (A), (B), (C) and (E) Only

(4) (A), (B), (C) and (D) Only

ANSWER: D

Let's evaluate each assumption of the Gaussian plume model for estimating pollutant concentration from a point source:

(A) **Concentrations are proportional to emission rate**:

- This is **true**. The model assumes that the concentration of pollutants is directly proportional to the emission rate.

(B) **Emissions take place from a virtual point**:

- This is **true**. The model often considers emissions originating from a virtual point source, especially for stacks and chimneys.

(C) **Pollutant does not undergo chemical transformations**:

- This is **true**. The model assumes that pollutants remain chemically inert during transport, meaning no chemical reactions occur.

(D) **Three-dimensional axis system of downwind, crosswind, and vertical with origin at the ground**:

- This is **true**. The model uses a three-dimensional coordinate system to describe pollutant dispersion, with axes representing downwind, crosswind, and vertical directions, and the origin usually placed at the source or the ground level.

(E) **Considers variable wind speed and direction**:

- This is **false**. The Gaussian plume model typically assumes steady-state conditions, meaning it does not account for variable wind speeds and directions.

Given this analysis, the correct answer is:

4. **(A), (B), (C) and (D) Only**

Which one of the following gases undergoes photo dissociation in troposphere?

(1) Carbonyl sulfur (COS)

(2) Nitrous Oxide (N_2O)

(3) Oxygen (O_2)

(4) Nitrogen dioxide (NO_2)

ANSWER: D

Let's dive into the details of photodissociation in the troposphere, particularly focusing on nitrogen dioxide (NO_2).

Photodissociation:

Photodissociation, also known as photolysis, is a chemical reaction in which a chemical compound is broken down by photons. It is a crucial process in atmospheric chemistry, as it initiates various photochemical reactions.

Nitrogen Dioxide (NO_2) Photodissociation:

Reaction:

In the troposphere, nitrogen dioxide (NO_2) undergoes photodissociation when exposed to sunlight. The reaction can be represented as:

$$\mathbf{NO_2 + hv \rightarrow NO + O}$$

Here, "hv" represents a photon of light.

Explanation:

1. **Absorption of Light**:
    - NO_2 molecules absorb photons from sunlight, especially ultraviolet (UV) light.
2. **Breaking of Bonds**:

- The energy from the absorbed photons breaks the chemical bonds in the NO_2 molecule, resulting in the formation of nitrogen monoxide (NO) and a free oxygen atom (O).

3. **Formation of Ozone (O_3)**:
    - The free oxygen atom (O) can react with molecular oxygen (O_2) to form ozone (O_3). This process is a key component of photochemical smog formation.

$$\mathbf{O + O_2 \rightarrow O_3}$$

Importance in Atmospheric Chemistry:

- **Photochemical Smog**:
    - Photodissociation of NO_2 is a critical step in the formation of ground-level ozone and other components of photochemical smog. This type of smog is common in urban areas with high vehicle emissions and industrial activities.
- **Air Quality and Health**:
    - Ground-level ozone formed through photodissociation is a major air pollutant that can cause respiratory problems, eye irritation, and other health issues. It also affects the environment by damaging crops and vegetation.

Other Gases in the Troposphere:

- **Carbonyl Sulfide (COS)**: Undergoes photodissociation in the stratosphere rather than the troposphere.
- **Nitrous Oxide (N_2O)**: Mainly photodissociates in the stratosphere, leading to the production of nitrogen oxides (NOx) that contribute to ozone depletion.
- **Oxygen (O_2)**: Primarily undergoes photodissociation in the stratosphere to form ozone through the ozone-oxygen cycle.

Which of the following is not related to incidents of deadly urban smogs?

(1) Meuse valley (Belgium), 1930

(2) Donora (Pennsylvania, USA), 1948

(3) London (UK), 1952

(4) Quebec (Canada), 1968

ANSWER: D

The correct answer is **(4) Quebec (Canada), 1968**.

The other three incidents—Meuse Valley in 1930, Donora in 1948, and London in 1952—are all well-known cases of deadly urban smog

Major Smog Incidents

Type of Smog	Incident Location	Year	Description
Sulfurous Smog	Meuse Valley, Belgium	1930	One of the earliest documented deadly smogs, caused by industrial emissions in a river valley.
Sulfurous Smog	Donora, Pennsylvania, USA	1948	A severe smog event caused by industrial pollution in a steel mill town, leading to several deaths.
Sulfurous Smog	London, UK	1952	Known as the "Great Smog," this event caused thousands of deaths due to high levels of coal smoke and sulfur dioxide.
Photochemical Smog	Los Angeles, USA	1940s-1950s	Frequent occurrences of photochemical smog due to vehicle emissions and sunny weather conditions.
Photochemical Smog	Mexico City, Mexico	1980s-1990s	Severe air pollution events caused by high vehicle emissions and geographic factors.
Photochemical Smog	Beijing, China	2000s-2010s	Widespread smog events due to rapid industrialization, vehicle emissions, and coal burning.

Which one of the following species in the atmosphere does NOT cause indirect radiative forcing?

(1) Aerosols

(2) Halocarbons

(3) Nitric oxide

(4) Ozone

ANSWER: D

Choose the correct statements:

(A) Radioactive decay is simplest example of true 1st order reaction

(B) It is customary to express decomposition rate of radioactive material in terms of half life

(C) Radioactive decay constant of an element varies significantly over wide environment conditions

(D) SI unit of radioactivity is curie (Ci)

(E) ß particles are actually electrons and ionising power is much weaker than a particles

Choose the correct answer from the options given below:

(1) (A), (B), (C), (D) and (E) Only

(2) (A), (C), (D) and (E) Only

(3) (A), (B) and (E) Only

(4) (A), (B), (C) and (D) Only

ANSWER: C

Let's review each statement:

(A) **Radioactive decay is simplest example of true 1st order reaction**:

- **True**. Radioactive decay follows first-order kinetics, meaning the rate of decay is directly proportional to the amount of the substance present.

(B) **It is customary to express decomposition rate of radioactive material in terms of half-life**:

- **True**. The half-life is a common way to express the rate of decay of radioactive materials.

(C) **Radioactive decay constant of an element varies significantly over wide environmental conditions**:

- **False**. The decay constant is a fixed property of a radioactive isotope and does not vary with environmental conditions.

(D) **SI unit of radioactivity is curie (Ci)**:

- **False**. The SI unit of radioactivity is the becquerel (Bq). The curie (Ci) is an older unit.

(E) **ß particles are actually electrons and ionising power is much weaker than α particles**:

- **True**. Beta particles (ß) are high-energy, high-speed electrons or positrons emitted by certain types of radioactive nuclei. They have weaker ionizing power compared to alpha (α) particles.

Given this analysis, the correct answer is:

3. **(A), (B) and (E) Only**

The upward movement of a parcel of air requires the mechanism, which may affect the ascent of air. Such mechanism is called lifting mechanism and falls into three categories which are:

A) The adiabatic lifting is caused due to cooling of a parcel of ascending air.

(B) The cyclonic lifting is caused due to meeting of two thermally contrasted air masses and subsequent formation of cyclonic front.

(C) The orographic lifting is caused by orographic barrier where in a horizontally moving air encounters hills or mountains and is forced to rise along the windward slope of the hills.

(D) The condensation lifting is caused due to decrease of temperature of a parcel of ascending air beyond condensation level.

(E) The convectional lifting is caused due to warming of surface air due to insolation heating. Choose the correct answer from the options given below :

Choose the correct answer from the options given below :

(1) (A), (B) and (D) Only

(2) (B), (C) and (E) Only

(3) (C), (D) and (E) Only

(4) (A), (D) and (E) Only

ANSWER: B

Let's evaluate each statement:

(A) **The adiabatic lifting is caused due to cooling of a parcel of ascending air.**

- This statement is **true**. Adiabatic lifting refers to the cooling of an air parcel as it ascends and expands due to lower pressure.

(B) **The cyclonic lifting is caused due to meeting of two thermally contrasted air masses and subsequent formation of cyclonic front.**

- This statement is **true**. Cyclonic lifting occurs when warm and cold air masses meet, creating a front that forces air to rise.

(C) **The orographic lifting is caused by orographic barrier where in a horizontally moving air encounters hills or mountains and is forced to rise along the windward slope of the hills.**

- This statement is **true**. Orographic lifting occurs when air is forced to ascend over a mountain range or other topographic barrier.

(D) **The condensation lifting is caused due to decrease of temperature of a parcel of ascending air beyond condensation level.**

- This statement is **false**. While condensation can occur when air cools to its dew point, it is not typically referred to as a distinct lifting mechanism.

(E) **The convectional lifting is caused due to warming of surface air due to insolation heating.**

- This statement is **true**. Convectional lifting occurs when the surface air is heated by solar radiation, causing it to become buoyant and rise.

Given this analysis, the correct answer is:

2. **(B), (C) and (E) Only**

Given below are two statements:

Statement (I): Special Vortex weather conditions in the lower stratosphere above Antarctica in winter causes denitrification and lead to conversion of inactive chlorine into active chlorine molecules

Statement (II): HCI and $CIONO_2$ are two important active chlorine compounds

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

ANSWER: C

Let's evaluate each statement:

Statement (I):

Special vortex weather conditions in the lower stratosphere above Antarctica in winter cause denitrification and lead to the conversion of inactive chlorine into active chlorine molecules.

- This statement is **correct**. During the Antarctic winter, the formation of a polar vortex leads to extremely cold temperatures in the lower stratosphere. These conditions cause the formation of polar stratospheric clouds (PSCs). Reactions on the surface of PSCs convert reservoir chlorine compounds into active chlorine species, such as Cl_2 and $ClONO_2$.

Statement (II):

HCl and $ClONO_2$ are two important active chlorine compounds.

- This statement is **incorrect**. HCl (hydrochloric acid) and $ClONO_2$ (chlorine nitrate) are reservoir or inactive forms of chlorine. The active forms of chlorine, which participate in ozone depletion, include Cl_2 (chlorine gas), ClO (chlorine monoxide), and HOCl (hypochlorous acid).

Given this analysis, the correct answer is:

3. **Statement I is correct but Statement II is incorrect**

Arrange the following Green house gases in their decreasing order in terms of the amount of extra warming they produce:

It should be increasing order

(A) CO2

(B) CH_4

(C) CFCs

(D) N_2O

(E) O_3

Choose the correct answer from the options given below :

(1) (A), (B), (E), (D), (C)

(2) (A), (B), (E), (C), (D)

(3) (A), (B), (D), (E), (C)

(4) (A), (E), (B), (D), (C)

ANSWER: B

According to NTA

Important Sequence :- For Global Warming Potential/Extra warming they Produce

$$\mathbf{SF_6 > N_2O > CFCs > O_3 > CH_4 > CO_2}$$

Atmospheric lifetime and global warming potential (GWP) relative to CO_2 at different time horizon for various greenhouse gases (more values provided at global warming potential)

Gas name	Chemical formula	Lifetime (years)	Radiative Efficiency ($Wm^{-2}ppb^{-1}$, molar basis)	Global warming potential (GWP) for given time horizon		
				20-yr.	100-yr.	500-yr.[7]:
Carbon dioxide	CO_2	50-200	1.37×10^{-5}	1	1	1
Methane (fossil)	CH_4	12	5.7×10^{-4}	83	30	10
Methane (non-fossil)	CH_4	12	5.7×10^{-4}	81	27	7.3
Nitrous oxide	N_2O	109	3×10^{-3}	273	273	130
CFC-11	CCl_3F	52	0.29	8321	6226	2093
CFC-12	CCl_2F_2	100	0.32	10800	10200	5200
HCFC-22	$CHClF_2$	12	0.21	5280	1760	549
HFC-32	CH_2F_2	5	0.11	2693	771	220
HFC-134a	CH_2FCF_3	14	0.17	4144	1526	436
Tetrafluoromethane	CF_4	50 000	0.09	5301	7380	10587
Hexafluoroethane	C_2F_6	10 000	0.25	8210	11100	18200
Sulfur hexafluoride	SF_6	3 200	0.57	17500	23500	32600
Nitrogen trifluoride	NF_3	500	0.20	12800	16100	20700

Ambient atmosphere have equal concentration of CO2, SO2, 03 and NO2 in the units of PPM. Which of the gas will have highest concentration in µg/m³ units at same temperature and pressure conditions ?

(1) CO_2

(2) SO_2

(3)O_3

(4) NO_2

ANSWER: B

To determine which gas will have the highest concentration in µg/m³ when their concentrations are equal in parts per million (ppm), we need to consider their molar masses. The gas with the highest molar mass will have the highest concentration in µg/m³, since µg/m³ depends on the mass of the gas molecules.

Here are the molar masses of the gases:

- **CO_2 (Carbon Dioxide)**: 44 g/mol
- **SO_2 (Sulfur Dioxide)**: 64 g/mol
- **O_3 (Ozone)**: 48 g/mol
- **NO_2 (Nitrogen Dioxide)**: 46 g/mol

Among these, **SO_2 (Sulfur Dioxide)** has the highest molar mass.

Therefore, the gas with the highest concentration in µg/m³, given equal concentrations in ppm, is:

2. **SO_2 (Sulfur Dioxide)**

Molar Mass of Gases:

- **CO_2 (Carbon Dioxide)**: 44 g/mol
- **SO_2 (Sulfur Dioxide)**: 64 g/mol
- **O_3 (Ozone)**: 48 g/mol
- **NO_2 (Nitrogen Dioxide)**: 46 g/mol

Step-by-Step Calculation:

1. **Concentration in µg/m³ for CO_2:**

$$\mu g/m^3 = 1 \text{ ppm} \times \frac{44 \text{ g/mol}}{22.414} \approx 1963 \ \mu g/m^3$$

2. **Concentration in µg/m³ for SO_2:**

$$\mu g/m^3 = 1 \text{ ppm} \times \frac{64 \text{ g/mol}}{22.414} \approx 2856 \ \mu g/m^3$$

3. **Concentration in µg/m³ for O_3:**

$$\mu g/m^3 = 1 \text{ ppm} \times \frac{48 \text{ g/mol}}{22.414} \approx 2141 \ \mu g/m^3$$

4. **Concentration in µg/m³ for NO_2:**

$$\mu g/m^3 = 1 \text{ ppm} \times \frac{46 \text{ g/mol}}{22.414} \approx 2051 \ \mu g/m^3$$

Summary of Results:

- **CO_2**: 1963 µg/m³
- **SO_2**: 2856 µg/m³
- **O_3**: 2141 µg/m³
- **NO_2**: 2051 µg/m³

JUNE 2023

Which one of the following gas was not released through volcanoes around 4 billion years ago?

1. Hydrogen
2. Nitrogen
3. Oxygen
4. Carbon dioxide

ANSWER: C

The gas that was not released through volcanoes around 4 billion years ago is:

3. **Oxygen**

Explanation:

- **Hydrogen, Nitrogen, and Carbon dioxide**: These gases were indeed released through volcanic activity in the early Earth's history. Volcanic eruptions contributed significantly to the primordial atmosphere with gases like water vapor (H_2O), carbon dioxide (CO_2), nitrogen (N_2), hydrogen (H_2), and various sulfur compounds.
- **Oxygen**: Molecular oxygen (O_2) was not released through volcanic activity. In fact, the early Earth's atmosphere had very little free oxygen. The significant increase in atmospheric oxygen occurred much later, around 2.4 billion years ago, due to the process of photosynthesis carried out by cyanobacteria during the Great Oxygenation Event.

Which of the following is not applicable to secondary air pollutants?

1. They are important with respect to human health.
2. They have undergone chemical/physical transformation in air.
3. They include both inorganic and organic species/compound.
4. they are non-hazardous air pollutants.

ANSWER: D

The statement that is not applicable to secondary air pollutants is:

4. **They are non-hazardous air pollutants.**

Explanation:

- **Secondary air pollutants** are formed in the atmosphere through chemical or physical transformations of primary pollutants.
    - Examples include ozone (O_3), which forms from the reaction of nitrogen oxides (NOx) and volatile organic compounds (VOCs) in the presence of sunlight, and sulfuric acid (H_2SO_4), which forms from sulfur dioxide (SO_2) reactions.
- **Health Impact**: Secondary air pollutants can have significant impacts on human health, causing respiratory and cardiovascular issues, among other health effects.
- **Composition**: They include both inorganic and organic species or compounds, resulting from complex reactions in the atmosphere.

Therefore, the statement that they are non-hazardous air pollutants is incorrect, as secondary air pollutants can be quite hazardous to health and the environment.

The increasing order of greenhouse gases in terms of the amount of extra warming they produce in earth's atmosphere is:

A. CO_2

B. CH_4

C. N_2O

D. CFCs

Choose the correct answer from the options given below:

1. A<B<C<D

2.D<C<B<A

3. C<D<B<A

4. C<B<D<A

ANSWER: C

According to NTA

Important Sequence :- For Global Warming Potential/Extra warming they Produce

$$\mathbf{SF_6 > N_2O > CFCs > O_3 > CH_4 > CO_2}$$

Given below are two statements:

Statement I: Process of incineration that destroys highly toxic and hazardous organic waste differs from municipal solid waste incineration where energy is often produced.

Statement II: Incineration process cannot be used with waste that have high concentration of water and non- combustible solids.

In the light of the above statements, choose the correct answer from the options

1. Both Statement I and Statement are true
2. Both Statement I and Statement II are false
3. Statement I is true but Statement II is false
4. Statement I is false but Statement II is true

ANSWER: A

Let's review each statement:

Statement I:

The process of incineration that destroys highly toxic and hazardous organic waste differs from municipal solid waste incineration where energy is often produced.

- **True**. Incineration of highly toxic and hazardous organic waste is specifically designed to ensure the complete destruction of harmful compounds, often using higher temperatures and additional controls to prevent the release of toxic emissions. In contrast, municipal solid waste incineration is often designed to recover energy, typically in the form of heat or electricity.

Statement II:

The incineration process cannot be used with waste that has a high concentration of water and non-combustible solids.

- **True**. Incineration is less effective for waste with high water content and non-combustible solids because these components do not burn and can significantly reduce the efficiency and effectiveness of the incineration process.

Given this analysis, the correct answer is:

1. **Both Statement I and Statement II are true**

Sound pressure is equal to:

1. Total atmospheric pressure only
2. Total Atmospheric Pressure minus Barometric pressure only
3. Barometric pressure only
4. Sum of total atmospheric pressure and barometric pressure

ANSWER: B

Which of the following is a catalytically inactive (or reservoir) molecule containing chlorine in stratosphere?

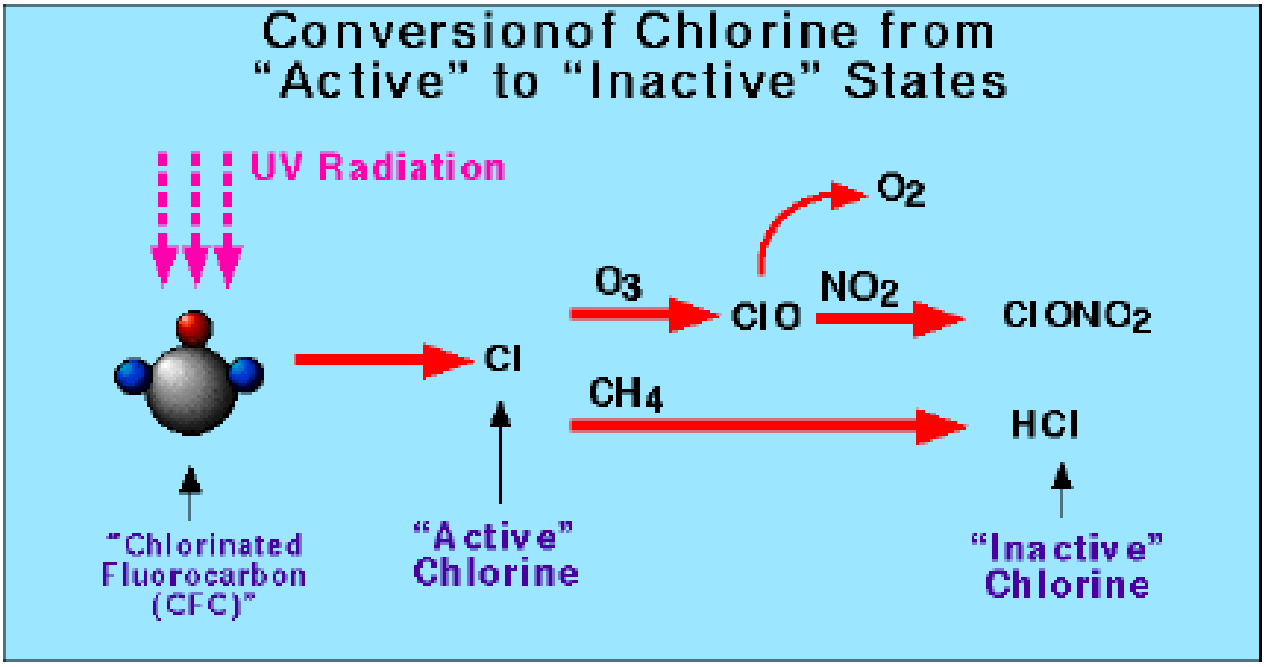
1. HOCl
2. ClO
3. $ClONO_2$
4. Cl_2

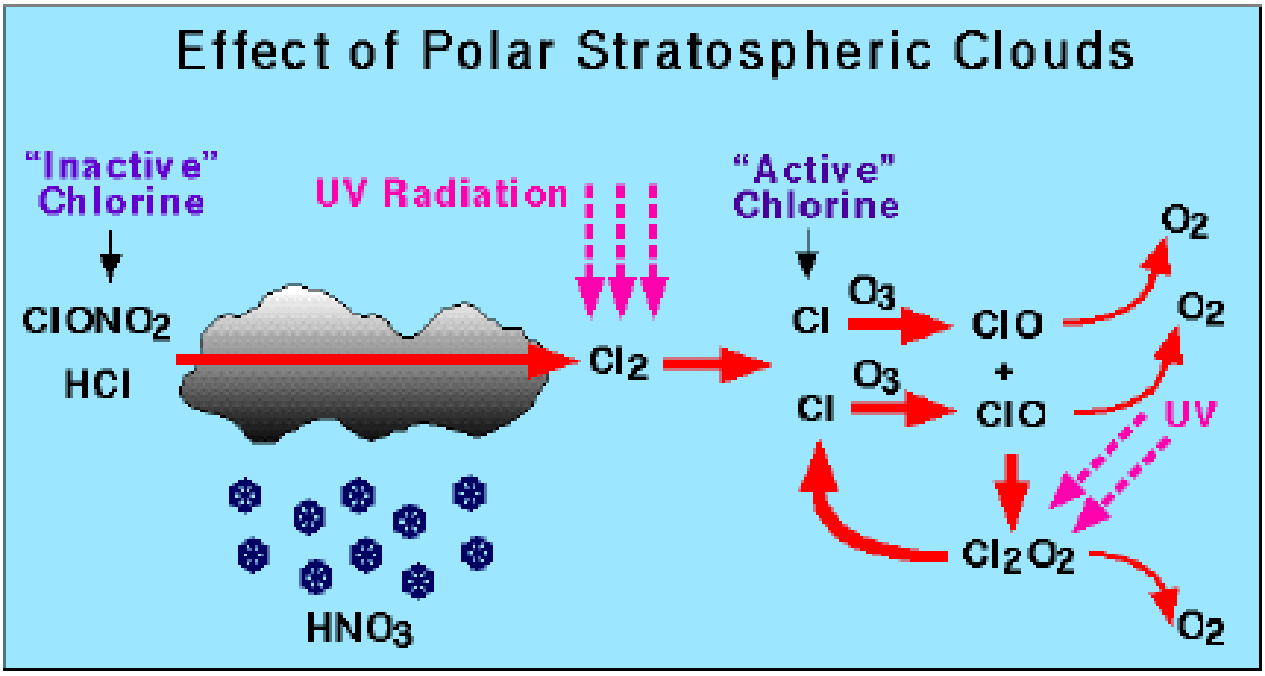
ANSWER: C

Molecule	Chemical Formula	Activity Status	Role in the Stratosphere
Chlorine Monoxide	ClO	Active	Participates in the catalytic destruction of ozone molecules.
Chlorine Nitrate	$ClONO_2$	Reservoir (Inactive)	Stores chlorine in an inactive form and can release active chlorine under certain conditions.
Chlorine Gas	Cl_2	Active/In Reservoir	Can be photolyzed to release active chlorine atoms, participating in ozone destruction.
Hypochlorous Acid	HOCl	Active	Participates in reactions that release active chlorine species, contributing to ozone depletion.

Summary:

- **ClO**: An active form of chlorine that directly destroys ozone.
- **$ClONO_2$**: A reservoir molecule that stores chlorine in an inactive form, which can release active chlorine species.
- **Cl_2**: Can act as a reservoir but is also readily photolyzed to release active chlorine.
- **HOCl**: An active chlorine compound that participates in ozone-depleting reactions.


Conversionof Chlorine from
"Active" to "Inactive" States
UV Radiation
O2
O3
ClO
NO2
ClONO2
Cl
CH4
HCl
"Chlorinated
Fluorocarbon
(CFC)"
"Active"
Chlorine
"Inactive"
Chlorine


Effect of Polar Stratospheric Clouds
"Inactive"
Chlorine
UV Radiation
"Active"
Chlorine
ClONO2
HCl
Cl2
Cl
O3
ClO
+
ClO
O2
O2
UV
Cl2O2
O2
HNO3

March 2023

An air parcel having initial temperature of 15°C rises adiabatically to 2 km height above the surface. What is the final temperature of the parcel?

1.~4.6°C

2.~5.6°C

3.~-4.6°C

4.~-6.4°C

ANSWER: C

To determine the final temperature of an air parcel rising adiabatically, we can use the dry adiabatic lapse rate (DALR), which is approximately 9.8°C per kilometer.

Steps to Calculate:

1. **Initial Temperature (T_0):** 15°C
2. **Height Change (Δh):** 2 km
3. **Dry Adiabatic Lapse Rate (DALR):** 9.8°C/km

The temperature change (ΔT) can be calculated using the formula:

$$\Delta T = \text{DALR} \times \Delta h$$

So,

$$\Delta T = 9.8°C/km \times 2 \text{ km}$$

$$\Delta T = 19.6°C$$

Since the temperature decreases as the parcel rises:

$$\text{Final Temperature } (T_1) = T_0 - \Delta T$$

$$T_1 = 15°C - 19.6°C$$

$$T_1 = -4.6°C$$

Therefore, the final temperature of the air parcel is:

3. **~-4.6°C**

Behaviour of the plume, when stack is under temperature inversion layer, is termed as

1. Fanning.
2. Conning.
3. Fumigation
4. Lofting.

ANSWER: C

The behavior of the plume when a stack is under a temperature inversion layer is termed as:

3. **Fumigation**

Explanation:

- **Temperature Inversion Layer**: This occurs when a layer of warmer air traps cooler air near the ground, preventing the upward movement of air and pollutants.
- **Fumigation**: In this scenario, the pollutants emitted from the stack are trapped below the inversion layer and are forced to spread horizontally. This can cause high concentrations of pollutants near the ground level.

Other plume behaviors under different conditions include:

- **Fanning**: Occurs during stable atmospheric conditions where the plume spreads horizontally.
- **Conning**: Refers to a vertically dispersed plume, typically under unstable atmospheric conditions.
- **Lofting**: Happens when the plume rises above an inversion layer and disperses in the upper atmosphere, avoiding the inversion layer.

The time between successive peaks or between successive troughs of the sinusoidal sound wave is called

1. Frequency (f)
2. Wavelength (λ)
3. Period (P)
4. Amplitude (A)

ANSWER: C

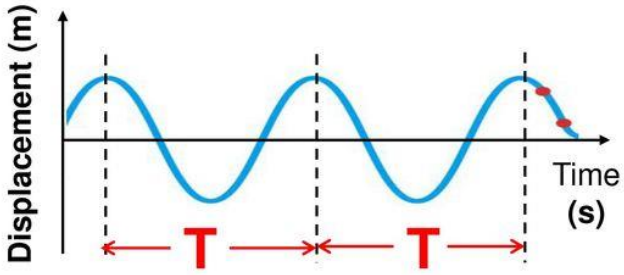
The time between successive peaks or between successive troughs of a sinusoidal sound wave is called:

3. **Period (P)**

Explanation:

- **Frequency (f)**: The number of cycles (or waves) that pass a given point per unit of time, typically measured in Hertz (Hz).
- **Wavelength (λ)**: The distance between successive peaks (or troughs) of the wave, typically measured in meters.
- **Period (P)**: The time it takes for one complete cycle of the wave to pass a given point, typically measured in seconds. It is the reciprocal of the frequency ($P = 1/f$).
- **Amplitude (A)**: The maximum displacement or distance moved by a point on the wave from its equilibrium position, typically measured in units of pressure or displacement.

PERIOD (T) *or* FREQUENCY (f)



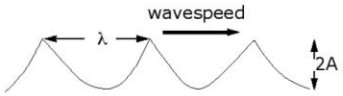
Period and frequency are inversely related

$$T = \frac{1}{f}$$

- An observer at a *fixed position* observes the wave moving by
- The time between successive crests passing by (or troughs) is the PERIOD
- The number of crests passing by per unit time is the FREQUENCY

Basic Wave Parameters

The amplitude A, is half the height difference between a peak and a trough.
The wavelength λ, is the distance between successive peaks (or troughs).
The period T, is the time between successive peaks (or troughs).
The wave speed c, is the speed at which peaks (or troughs) move.
The frequency ν, (Greek letter "nu") measures the number of peaks (or troughs) that pass per second.
Note that "c" (from the Latin word "celeritas" meaning swiftness) is used for wave speed, not "v". One reason is so that it is not confused with frequency.



$$\text{Wave speed} = \frac{\text{distance between peaks}}{\text{time between peaks}}$$

$$c = \frac{\lambda}{T} = \nu\lambda$$

Choose the correct statements

A. VOC, NOx and sunlight are essential for photochemical smog.

B. Photochemical smog is predominantly formed during the summer season.

C. NO_2 dissociates by sunlight at 300-400 nm.

D. Molecular O_2 dissociate by sunlight and provide atomic O for O_3 formation in troposphere.

E. Tropospheric O_3 concentration is maximum at noon on a given day.

Choose the correct answer from the options given below:

1. A. D and E only
2. A, B, C and E only.
3. A, B, D and E only
4. B. C, D and E only

ANSWER: B

Let's analyze each statement to determine the correct answer:

A. **VOC, NOx and sunlight are essential for photochemical smog.**

- **True**. Volatile Organic Compounds (VOCs), nitrogen oxides (NOx), and sunlight are crucial components in the formation of photochemical smog.

B. **Photochemical smog is predominantly formed during the summer season.**

- **True**. Photochemical smog is more common in the summer due to higher temperatures and stronger sunlight, which enhance the chemical reactions.

C. **NO2 dissociates by sunlight at 300-400 nm.**

- **True**. Nitrogen dioxide (NO_2) dissociates under sunlight in the 300-400 nm range, releasing atomic oxygen which can further react to form ozone (O_3).

D. **Molecular O2 dissociate by sunlight and provide atomic O for O3 formation in the troposphere.**

- **False**. Molecular oxygen (O_2) primarily dissociates in the stratosphere to form ozone, not in the troposphere. In the troposphere, ozone formation mainly involves the reaction of NO_2 and sunlight.

E. **Tropospheric O3 concentration is maximum at noon on a given day.**

- **True**. Tropospheric ozone concentration tends to peak around noon or early afternoon when sunlight is strongest.

Given this analysis, the correct answer is:

2. **A, B, C and E only**

Choose the correct statement

A. UV-C breaks down O_2 molecule in the stratosphere.

B. A 1% decrease in the overhead O_3 in the stratosphere results in 2% increase in UV-B intensity at the ground level.

C. UV-A is biologically, the least harmful.

D. UV-B helps in O_3 dissociation in the stratosphere.

E. Chapman mechanism deals with both, O_3 formation and destruction, in the stratosphere.

Choose the correct answer from the options given below:

1. A, B, C and D only

2. A, C, D and E only

3. A, B, C, D and E

4. A, B, D and E only

ANSWER: C

Let's analyze each statement:

A. **UV-C breaks down O_2 molecule in the stratosphere.**

- **True**. UV-C radiation (wavelengths less than 280 nm) has enough energy to break the bond in an O_2 molecule, leading to the formation of two oxygen atoms.

B. **A 1% decrease in the overhead O_3 in the stratosphere results in 2% increase in UV-B intensity at the ground level.**

- **True**. A decrease in stratospheric ozone allows more UV-B radiation to reach the ground. The approximation is that for every 1% decrease in ozone, there is about a 2% increase in UV-B intensity.

C. **UV-A is biologically, the least harmful.**

- **True**. Among the different types of UV radiation, UV-A (320-400 nm) is the least harmful to biological organisms compared to UV-B and UV-C.

D. **UV-B helps in O_3 dissociation in the stratosphere.**

- **True**. The dissociation of ozone is mainly due to UV-C and UV-B radiation.

E. **Chapman mechanism deals with both, O_3 formation and destruction, in the stratosphere.**

- **True**. The Chapman mechanism describes the processes of ozone formation and destruction in the stratosphere, involving UV radiation and various oxygen species.

Given this analysis, the correct answer is:

2. **A, B, C, D and E**

The international stable isotope standard Vienna Peedee Belemnite (VPDB) is used for the following stable isotope systems:

A. $_{34}S/_{32}S$

B. 2H/H

C. $_{18}O/_{16}O$

D. $_{16}N/_{15}N$

E. $_{13}C/_{12}C$

Choose the correct answer from the options given below:

1. A and B only

2. B and C only

3. C and E only

4. D and E only

ANSWER: C

The Vienna Pee Dee Belemnite (VPDB) standard is a reference material used for stable isotope measurements of carbon (13C/12C) and oxygen (18O/16O) isotopes. It is named after the original Pee Dee Belemnite (PDB), a Cretaceous marine fossil from the Pee Dee formation in South Carolina, USA.

Key Points:

- **Purpose**: VPDB serves as an internationally recognized standard for comparing isotopic compositions of various samples in geochemistry, paleoclimatology, and other fields.
- **Isotopes Measured**:
    - **Carbon Isotopes (13C/12C)**: Used in studies of carbon cycles, paleoclimates, and ecological research.
    - **Oxygen Isotopes (18O/16O)**: Used to analyze water cycles, paleotemperatures, and climatic changes.

The variability in height of tropopause is dependent on following factors

A. Longitude

B. Latitude

C. Temperature

D. Season

E. Vegetation

Choose the correct answer from the options given below:

1. A, B and C only

2. B, C and D only

3. A, C and E only

4. A, B, and only

ANSWER: B

The height of the tropopause, which is the boundary between the troposphere and the stratosphere, varies depending on several key factors. Let's discuss these factors in detail:

1. Latitude

- **Equator**: The tropopause is higher at the equator, typically around 16 to 18 kilometers (10 to 11 miles). This is due to the warm temperatures and the intense upward convection of air in tropical regions.
- **Poles**: The tropopause is lower at the poles, generally around 8 to 10 kilometers (5 to 6 miles). This is because of the colder temperatures and less vigorous vertical air movement in polar regions.
- **Mid-Latitudes**: The height of the tropopause in mid-latitude regions is intermediate, often around 10 to 12 kilometers (6 to 7.5 miles).

2. Temperature

- The temperature profile of the atmosphere affects the height of the tropopause.
- **Higher Temperatures**: In warmer regions, the tropopause is higher because warm air rises more vigorously, pushing the boundary between the troposphere and the stratosphere upwards.
- **Lower Temperatures**: In colder regions, the tropopause is lower because there is less vertical movement of air.

3. Season

- **Summer**: During the summer, the tropopause is higher due to increased solar heating, which causes more intense upward convection of air.
- **Winter**: In winter, the tropopause is lower because of reduced solar heating and weaker vertical air movement.

Factor	Effect on Tropopause Height
Latitude	Higher at the equator, lower at the poles, intermediate at mid-latitudes
Temperature	Higher in warmer regions, lower in colder regions
Season	Higher in summer, lower in winter
Longitude	No significant impact
Vegetation	No direct impact

Criteria pollutant under National Ambient Air Quality Standards (NAAQS) are

A. CO

B. CO_2

C. O_3

D. SO_2

E. NO_3

Choose the correct answer from the options given below:

1. A, C, D and E only
2. A, C and D only
3. A, B and D only
4. A, D and E only

ANSWER: B

The criteria pollutants under the National Ambient Air Quality Standards (NAAQS) include:

- **CO (Carbon Monoxide)**
- **O_3 (Ozone)**
- **SO_2 (Sulfur Dioxide)**

The correct answer is:

2. **A, C and D only**

Explanation:

- **CO (Carbon Monoxide)**: A criteria pollutant regulated due to its harmful effects on human health.
- **CO_2 (Carbon Dioxide)**: Not a criteria pollutant under NAAQS. It is a greenhouse gas but not classified as a criteria pollutant for air quality standards.
- **O_3 (Ozone)**: A criteria pollutant regulated due to its role in smog formation and harmful health effects.
- **SO_2 (Sulfur Dioxide)**: A criteria pollutant regulated due to its impact on respiratory health and the environment.
- **NO_3 (Nitrate)**: Not a criteria pollutant under NAAQS.

Criteria Air Pollutants

EPA (Environmental Protection Agency), USA has set national air quality standards for sic pollutants known as six criteria pollutants:

1. Ground-level ozone (O_3)
2. Carbon monoxide (CO)
3. Sulfur dioxide (SO_2)
4. Small particulates PM10 and now PM2.5
5. Nitrogen dioxide (NO_2)
6. Lead (Pb).

Revised National Ambient Air Quality Standards (NAAQS)

[NAAQS Notification dated 18th November, 2009]

S. No.	Pollutants	Time Weighted Average	Concentration in Ambient Air		Methods of Measurement
			Industrial, Residential, Rural and other Areas	Ecologically Sensitive Area (notified by Central Government)	
1	Sulphur Dioxide (SO_2), µg/m^3	Annual*	50	20	1. Improved West and Gaeke 2. Ultraviolet Fluorescence
		24 Hours**	80	80	
2	Nitrogen Dioxide (NO_2), µg/m^3	Annual*	40	30	1. Modified Jacob & Hochheiser 2. Chemiluminescence
		24 Hours**	80	80	
3	Particulate Matter (Size <10µm) or PM_{10} µg/m^3	Annual*	60	60	1. Gravimetric 2. TEOM 3. Beta attenuation
		24 Hours**	100	100	
4	Particulate Matter (Size <2.5 µm) or $PM_{2.5}$ µg/m^3	Annual*	40	40	1. Gravimetric 2. TEOM 3. Beta attenuation
		24 Hours **	60	60	
5	Ozone (O_3), µg/m^3	8 hours**	100	100	1. UV photometric 2. Chemiluminescence 3. Chemical Method
		1 hours **	180	180	
6	Lead (Pb), µg/m^3	Annual *	0.50	0.50	1. AAS/ICP Method after sampling using EPM 2000 or equivalent filter paper 2. ED-XRF using Teflon filter
		24 Hour**	1.0	1.0	
7	Carbon Monoxide (CO), mg/m^3	8 Hours **	02	02	Non dispersive Infra Red (NDIR) Spectroscopy
		1 Hour**	04	04	
8	Ammonia (NH_3), µg/m^3	Annual*	100	100	1. Chemiluminescence 2. Indophernol blue method
		24 Hour**	400	400	
9	Benzene (C_6H_6) , µg/m^3	Annual *	05	05	1. Gas chromatography based continuous analyzer 2. Adsorption and Desorption followed by GC analysis
10	Benzo(a)Pyrene (BaP)-particulate phase only, ng/m^3	Annual*	01	01	Solvent extraction followed by HPLC/GC analysis
11	Arsenic (As), ng/m^3	Annual*	06	06	AAS/ICP method after sampling on EPM 2000 or equivalent filter paper
12	Nickel (Ni), ng/m^3	Annual*	20	20	AAS/ICP method after sampling on EPM 2000 or equivalent filter paper

** Annual Arithmetic mean of minimum 104 measurements in a year at a particular site taken twice a week 24 hourly at uniform interval.** 24 hourly 08 hourly or 01 hourly monitored values, as applicable shall be complied with 98% of the time in a year. 2% of the time, they may exceed the limits but not on two consecutive days of monitoring.*

NOTE: Whenever and wherever monitoring results on two consecutive days of monitoring exceed the limits specified above for the respective category, it shall be considered adequate reason to institute regular or continuous monitoring and further investigation.

Air pollution control techniques used primarily to control particulate matter pollution in ambient atmosphere are

A. Bag houses

B. Limestone injection multistage burner (LIMB)

C. Cyclone collectors

D. Catalytic converters

E. Electrostatic precipitators

Choose the correct answer from the options given below:

1. A, B, D and E only

2. C and E only

3. A, B, C and E only

4. A, C and E only

ANSWER: D

The air pollution control techniques used primarily to control particulate matter pollution in the ambient atmosphere are:

A. **Bag houses**: These are fabric filter systems that trap particulate matter from the air. C. **Cyclone collectors**: These use centrifugal force to separate particulate matter from the air. E. **Electrostatic precipitators**: These use electrical charges to remove particulate matter from the air stream.

B. **Limestone injection multistage burner (LIMB)**: This is primarily used for controlling sulfur dioxide emissions rather than particulate matter. D. **Catalytic converters**: These are primarily used for controlling gaseous pollutants from vehicle exhausts, such as carbon monoxide, hydrocarbons, and nitrogen oxides, rather than particulate matter.

Given this analysis, the correct answer is:

4. **A, C and E only**

SEPTEMBER 2022

Primary air pollutants differ from secondary air pollutants as they

1. are present in levels that can be weighed on filter paper.

2. are directly emitted from identified sources.

3. have undergone chemical transformation in air after release from identified sources.

4. are designated as hazardous air pollutants.

ANSWER: B

Primary air pollutants differ from secondary air pollutants in that they:

2. **are directly emitted from identified sources.**

Explanation:

- **Primary Air Pollutants**: These are pollutants that are emitted directly from a source. Examples include carbon monoxide (CO) from vehicle exhausts, sulfur dioxide (SO_2) from power plants, and particulate matter from industrial processes.
- **Secondary Air Pollutants**: These pollutants are not directly emitted but form in the atmosphere through chemical or physical transformations of primary pollutants. Examples include ozone (O_3), which forms from the reaction of nitrogen oxides (NOx) and volatile organic compounds (VOCs) in the presence of sunlight, and sulfuric acid (H_2SO_4), which forms from sulfur dioxide (SO_2) reactions.

What will be the concentration of 40 ppb SO2 in µg/m³ unit at 25°C?

1. ~0.04 µg/m³
2. ~1047 µg/m³
3. ~104.7 µg/m³
4. ~10.47 µg/m³

ANSWER: C

To convert the concentration of SO_2 from ppb (parts per billion) to µg/m³ (micrograms per cubic meter), we can use the following formula:

$$\text{Concentration in } \mu g/m^3 = \text{Concentration in ppb} \times \frac{\text{Molecular Weight of } SO_2}{24.45}$$

Where 24.45 is the molar volume of air in liters at standard temperature and pressure (STP).

Step-by-Step Calculation:

1. **Given:**
   - Concentration = 40 ppb
   - Molecular weight of SO_2 = 64.07 g/mol
   - Temperature = 25°C (which approximates to standard conditions for the purposes of this calculation)
2. **Calculate:**

$$\text{Concentration in } \mu g/m^3 = 40 \times \frac{64.07}{24.45}$$

$$\text{Concentration in } \mu g/m^3 \approx 40 \times 2.62$$

$$\text{Concentration in } \mu g/m^3 \approx 104.7$$

Therefore, the concentration of 40 ppb SO_2 in µg/m³ is approximately:

3. **~104.7 µg/m³**

Given below are two statements

Statement I: Halons used as fire extinguishers are bromine containing hydrogen free substances and are involved in catalytic destruction of ozone in Stratosphere.

Statement II: In order to protect ozone layer in stratosphere, use of CFC was globally replaced by HCFC's, HFC's followed by Halons after Kyoto protocol.

In light of the above statements, choose the correct answer from the options given below

1. Both Statement I and Statement II are true
2. Both Statement I and Statement II are false
3. Statement I is true but Statement II is false
4. Statement I is false but Statement II is true

ANSWER: C

Statement I:

Halons used as fire extinguishers are bromine-containing hydrogen-free substances and are involved in catalytic destruction of ozone in the stratosphere.

- **True**. Halons are indeed bromine-containing compounds, and they do not contain hydrogen. They are effective fire extinguishers but are also potent ozone-depleting substances. In the stratosphere, bromine atoms released from halons participate in catalytic cycles that destroy ozone.

Statement II:

In order to protect the ozone layer in the stratosphere, the use of CFCs was globally replaced by HCFCs, HFCs, followed by Halons after the Kyoto Protocol.

- **False**. The phase-out of CFCs was addressed by the Montreal Protocol, not the Kyoto Protocol. The Montreal Protocol aimed to protect the ozone layer by phasing out the production of numerous substances that are responsible for ozone depletion, including CFCs and halons. HCFCs (Hydrochlorofluorocarbons) and HFCs (Hydrofluorocarbons) were introduced as transitional substitutes for CFCs, with HCFCs having a lower ozone-depleting potential than CFCs, and HFCs having no ozone-depleting potential. The Kyoto Protocol, on the other hand, primarily addresses greenhouse gases and climate change.

Given this analysis, the correct answer is:

3. **Statement I is true but Statement II is false**

Match List I with List II

LIST-I Green House Gas		LIST-II Global Warming Potential (GWP) (in 20yrs time period)	
A.	CO_2	**I.**	360
B.	CH_4	**II.**	50
C.	N_2O	**III.**	280
D.	CH_3CCI_3	**IV.**	1

Choose the correct answer from the options given below:

1. A-II, B-I, C-III, D-IV

2. A-I, B-II, C-IV, D-III

3. A-III, B-II, C-I, D-IV

4. A-IV, B-II, C-III, D-I

ANSWER: D

- **CO_2 (Carbon Dioxide)**: The reference gas with a GWP of 1.
- **CH_4 (Methane)**: Has a GWP of approximately 50 over a 20-year period.
- **N_2O (Nitrous Oxide)**: Has a GWP of approximately 280 over a 20-year period.
- **CH_3CCl_3 (Methyl Chloroform)**: Has a GWP of approximately 360 over a 20-year period.

Matching these to List II:

- A. CO_2 (IV) - 1
- B. CH_4 (II) - 50
- C. N_2O (III) - 280
- D. CH_3CCl_3 (I) - 360

So the correct answer is:

4. **A-IV, B-II, C-III, D-I**

The sum of diffused and direct solar radiation, is called

1. continental solar radiation.
2. global solar radiation.
3. total solar radiation.
4. universal solar radiation.

ANSWER: B

The correct term for the sum of diffused and direct solar radiation is:

2. **global solar radiation**

Explanation:

- **Direct Solar Radiation**: The sunlight that reaches the Earth's surface without being scattered or diffused.
- **Diffused Solar Radiation**: The sunlight that has been scattered by molecules and particles in the atmosphere and reaches the Earth's surface indirectly.
- **Global Solar Radiation**: The total amount of solar radiation reaching the Earth's surface, which includes both direct and diffused solar radiation.

Given below are two statements about Baghouse filter used to control air pollution

Statement I: Baghouse filters are fabric bag filter which can be repeatedly cleaned and reused. Collection efficiency can be very high (99.9% for all size of particles).

Statement II: Collection efficiency of baghouse filters decreases as dust collected on filters and lead to decrease in pressure drop which increases airflow or decreases cost of forcing air through filters.

In light of the above statements, choose the correct answer from the options given below

1. Both Statement I and Statement II are true
2. Both Statement I and Statement II are false
3. Statement I is true but Statement II is false
4. Statement I is false but Statement II is true

ANSWER: C

Statement I:

Baghouse filters are fabric bag filters which can be repeatedly cleaned and reused. Collection efficiency can be very high (99.9% for all sizes of particles).

- **True**. Baghouse filters use fabric bags that can be cleaned and reused. They are highly efficient, with collection efficiencies of up to 99.9% for particles of various sizes.

Statement II:

Collection efficiency of baghouse filters decreases as dust collects on filters and leads to a decrease in pressure drop which increases airflow or decreases the cost of forcing air through filters.

- **False**. As dust collects on the filters, the collection efficiency typically increases because the dust layer itself acts as an additional filter. However, this also leads to an increase in pressure drop, not a decrease, which makes it more difficult to force air through the filters and can increase operational costs.

Given this analysis, the correct answer is:

3. **Statement I is true but Statement II is false**

National Ambient air quality standards typically include following components

1. Maximum and Minimum pollutant levels, adjustment for co-pollutants
2. Abetment strategies, non-attainment penalties
3. Ambient temperature, size distribution and relative humidity
4. Indicator (Chemical/Chemical class), Averaging time, Pollutant form and level

ANSWER: D

The correct components typically included in National Ambient Air Quality Standards (NAAQS) are:

4. **Indicator (Chemical/Chemical class), Averaging time, Pollutant form and level**

Explanation:

- **Indicator (Chemical/Chemical class)**: Specifies the particular pollutant being measured (e.g., CO, SO_2, $PM_{2.5}$).
- **Averaging time**: Defines the period over which the pollutant concentration is averaged (e.g., 1-hour, 8-hour, annual).
- **Pollutant form**: Describes the specific form of the pollutant being measured (e.g., ozone as O_3, particulate matter as PM_{10} or $PM_{2.5}$).
- **Pollutant level**: Sets the allowable concentration of the pollutant in the ambient air (e.g., 100 µg/m³).

Given below are two statements

Statement I: Noise pollution exists in the form of energy and is highly subjective in nature unlike air pollution.

Statement II: Noise pollution exists irrespective of loudness of sound and causes both physiological and psychological stress among individual or group of individuals.

In light of the above statements, choose the correct answer from the options given below

1. Both Statement I and Statement II are true
2. Both Statement I and Statement II are false
3. Statement I is true but Statement II is false
4. Statement I is false but Statement II is true

ANSWER: A

Let's analyze each statement:

Statement I:

Noise pollution exists in the form of energy and is highly subjective in nature unlike air pollution.

- **True**. Noise pollution is a form of sound energy that can vary greatly in perception from person to person, making it highly subjective. Different individuals might experience the same noise level differently depending on their sensitivity and context.

Statement II:

Noise pollution exists irrespective of the loudness of sound and causes both physiological and psychological stress among individuals or groups of individuals.

- **False**. Noise pollution is typically associated with loud or unwanted sounds. While it is true that noise pollution can cause both physiological and psychological stress, the statement that it exists irrespective of loudness is inaccurate. The loudness of sound is a crucial factor in defining noise pollution.

Given this analysis, the correct answer is:

3. **Statement I is true but Statement II is false**